TFPT: Carrier Algebra, Compression Identities, and Finite Completion

An overdetermined rank-5 spectral theory on the oriented double cover

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Abstract

A minimal carrier $E_3 \oplus E_2$ on the oriented double cover is shown to be the unique rank-5 split satisfying $\dim \Lambda^{\text{even}} E = 16$ and $\mathrm{Tr}_E\, Y^2 = 5/6$. This carrier already forces the global Standard Model group $S(U(3)\times U(2)) \cong (SU(3)\times SU(2)\times U(1)_Y)/\mathbb{Z}_6$, one chiral family with ν^c from the positive half-spinor, the hypercharge–seam inversion $6Y^2 - Y - \mathbf{1} = 0$, and a self-consistent electromagnetic closure equation whose positive root gives $\alpha^{-1}(0) = 137.036\,000\,0$.

On the canonical branch the carrier packet is overdetermined by a set of compression identities: the rank-lock $2^{g-3} = g - 1$ has the unique integral solution $g = 5$; the 41-chain $48\, I_1^{\text{EW}} = 32\, k_{\text{EW}} = 10\, b_1 = \Omega_{\text{adm}}\gamma + 1 = 41$ collapses four algebraic layers onto one integer; and the Cabibbo inversion $\varphi_0 = \frac{1}{2}(1 - \sqrt{1 - 4\lambda_C^2})$ compresses the bridge outputs $(\beta, \Omega_b, \lambda_C, \sin^2\theta_{13})$ into a single seed. Wrong discrete worlds fail the electromagnetic closure categorically: two families give $\alpha^{-1} \approx 156$, four families give $\alpha^{-1} \approx 125$, and two Higgs doublets give $\alpha^{-1} \approx 136$.

The remaining open problem is a finite completion program with five explicit interfaces: a family local-system theorem hardening the flavor lift and the occupancy interpretation of $\Omega_{\text{adm}} = 48$, a bosonic relative index selecting the unique light Higgs doublet, a χ-channel heat-kernel closure generating $\bar{M}_{\text{Pl}}$ and the electroweak vacuum from one geometric response, a discrete hexagonal transport kernel for Yukawas, mixings, confinement, and strong CP, and a comparison bridge fixing scheme and threshold conventions. E_8 is retained only as downstream scale grammar; record algebra, horizons, and prediction semantics are conditional appendix material. Benchmark tables, a falsification matrix, and finite proof obligations separate hard results from bridges and completion targets throughout.

Reading Status for the Present Paper

- **Hard:** oriented double cover, seam Σ, involution ι, minimal carrier $E_3 \oplus E_2$, global group $S(U(3) \times U(2)) \cong \frac{SU(3)\times SU(2)\times U(1)_Y}{\mathbb{Z}_6}$, one chiral family, carrier norm 5/6, the abelian-index formula, and the established kernel numerics $(c_3, \delta_{\text{top}}, \varphi_0, \alpha)$.
- **Bridge:** family space F, occupancy $\Omega_{\text{adm}} = 48$, the occupancy reinterpretation $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$, the comparison map $\mathcal{R}_{\text{SM}}$, and the benchmark interface from kernel outputs to observables.
- **Completion:** bosonic relative index for Higgs selection, χ-channel heat-kernel closure, regularized Yukawa kernels and path-length masses, confinement as admissible word selection, and strong-CP cancellation.
- **Conditional:** downstream E_8 scale grammar, record algebra, horizons, prediction semantics, and the final compiler theorem.

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1 Reading guide and claim map

The paper is organized in three reading layers:

- **Big picture:** one section explains what TFPT claims in plain language before the detailed datum and conventions begin.
- **Detail:** the main body then runs from primitive core and hard carrier kernel through kernel numerics, the two completion programs, the particle/mass compiler, and the benchmark interface.
- **Proof sketches:** the main text gives only the central proof sketches, while dense technical derivations remain in appendices or companion material.

The status language is also fixed once and for all:

- **Hard** means established carrier or kernel structure.
- **Bridge** means a structural lift or comparison interface that organizes the compiler but is not yet a closed theorem.
- **Completion** means a theorem-sized open interface that would close a genuine physics sector.

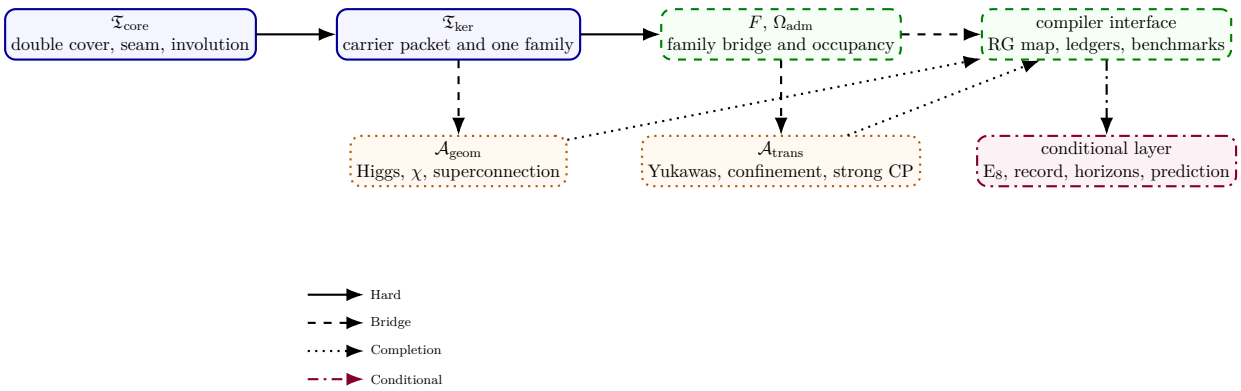

Figure 1: Claim map for the carrier-first compiler form. Hard fixes the carrier and numerical kernel, Bridge organizes family lift and comparison, Completion closes geometry and transport, and Conditional collects downstream extensions.

- **Conditional** means a downstream or appendix-level statement that is only accepted once the completion layer is granted.

2 Big picture: what TFPT actually claims

TFPT is presented here as a carrier-first theory of admissible vacua on the oriented double cover. The seam involution enforces the minimal carrier $E_3 \oplus E_2$. That carrier fixes the global Standard Model group, hypercharge, and one chiral family. The family count $N_{\rm fam} = 3$ is then supported by two independent routes: a local patch-deficit count and a global monodromy argument. The monodromy space F is retained not as the origin of the number three, but as the carrier of nondegeneracy, flavor phases, and mixing. A bosonic relative index is intended to select the Higgs, the χ channel is intended to generate both the Planck scale and the electroweak vacuum, and a discrete phase transport is intended to generate Yukawas, mixings, confinement, and strong CP. The comparison map then translates this compiler into PDG-style observables.

Carrier-first compiler chain

$$\begin{aligned}(\widetilde{M} \to M, \Sigma, \iota, \chi) &\Rightarrow (E_3 \oplus E_2,\ G_{\rm SM}^{\rm glob},\ S^+)\\ &\Rightarrow (N_{\rm fam} = 3,\ \Omega_{\rm adm} = 48,\ \Phi)\\ &\Rightarrow (\bar{M}_{\rm Pl},\ v,\ \mathcal{Y}_f)\\ &\Rightarrow (m_f,\ V_{\rm CKM},\ U_{\rm PMNS},\ \text{hadrons},\ \text{Pred}).\end{aligned} \tag{1}$$

The remainder of the main text sharpens each arrow of this chain in turn.

3 Primitive core and hard carrier kernel

3.1 Primitive core and conventions

We use natural units $c = \hbar = 1$. Relative operators are formulated in Euclidean signature and compared through a declared Wick-rotation bridge when Lorentzian language is needed. Higgs fields are denoted by Φ, Hamiltonians by $\mathcal{H}$, and the cosmological Hubble rate by $H_{\rm hub}(t)$; this avoids overloading the letter H. Newton's constant is always G_N, while $G_{\rm SM}^{\rm glob}$ is reserved for the global Standard Model gauge quotient. The seam is denoted by Σ, the involution by ι, and word parity by $\varepsilon(w)$.

Layer or statement	Status	Proof source or missing lemma	Main empirical or structural test
Oriented double cover, seam, involution	Hard	starting geometric datum; fixed throughout the paper	structural consistency of all later layers
Minimal carrier $E_3 \oplus E_2$	Hard	proof sketch given in Section 3.2	failure would collapse the whole packet
One-family packet $S^+ = \Lambda^{\text{even}} E$	Hard	decomposition and proof sketch given in the main text	group/matter mismatch would be global
Kernel numerics $(c_3, \delta_{\text{top}}, \varphi_0, \alpha)$	Hard	carrier-form closure equations summarized in the main text	scheme discipline for observable comparison
Family count $N_{\text{fam}} = 3$ from patch deficit	Hard	patch-deficit route and proof sketch given	family-count interpretation
Family space F for flavor splitting	Bridge	monodromy for nondegeneracy and mixing still open	flavor phases, CKM / PMNS
Occupancy $\Omega_{\text{adm}} = 48$ and $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$	Bridge	immediate count corollary; occupancy reinterpretation is bridge	topological surcharge
Comparison map $\mathcal{R}_{\text{SM}}$	Bridge	declared RG / threshold bookkeeping	convention-complete theory-versus-data translation
Bosonic relative index selecting Φ	Completion	missing bosonic index closure	unique light doublet / doublet-triplet split
χ-channel Einstein bridge	Completion	missing relative heat-kernel closure	one closure for $\bar{M}_{\text{Pl}}$, v, and the scale map
Regularized Yukawa kernel closure	Completion	missing phase-distance theorem	m_f, λ_C, θ_{13}, CKM / PMNS sharpenings
Confinement and strong CP closure	Completion	missing bound-state control and positivity hardening	hadron spectrum and $\bar{\theta}$ null tests
E_8, record, horizons, prediction semantics	Conditional	deferred to appendices and the conditional compiler theorem	downstream extension rather than main-text hard claim

Table 1: Status matrix for the carrier-first compiler form.

When color-center notation is needed later, we use multiplicative notation

$$z(w) \in \{1, \omega, \omega^2\}, \qquad \omega = e^{2\pi i/3}.$$

Thus center neutrality means $z(w) = 1$.

$$\mathfrak{T}_{\text{core}} := (\widetilde{M} \to M,\ \Sigma,\ \iota,\ D_{\text{rel}},\ \mathsf{Adm}).$$

Here "relative" always means comparison with a declared reference operator or reference-subtracted action, and Adm denotes the admissibility predicate on relative sectors. The technical tables for relative objects, layered bridge/completion data, and the constants atlas are collected in Appendix A and Appendix B; they are kept out of the early main-text flow so that the carrier kernel appears before the technical bookkeeping apparatus.

3.2 Minimal carrier theorem

Theorem 3.1 (Minimal carrier theorem)**.** *Let $E = E_b \oplus E_s$ be a split carrier with traceless diagonal generator*

$$Y_{b,s} = \operatorname{diag}\left(\underbrace{-\frac{1}{b}, \ldots, -\frac{1}{b}}_{b \text{ times}}, \underbrace{\frac{1}{s}, \ldots, \frac{1}{s}}_{s \text{ times}} \right).$$

If one simultaneously requires

$$\dim \Lambda^{\text{even}} E = 16, \qquad \operatorname{Tr}_E Y_{b,s}^2 = \frac{5}{6}, \tag{2}$$

then necessarily $\{b, s\} = \{3, 2\}$.

Proof. Write $g = b + s$. Since

$$\dim \Lambda^{\text{even}} E = 2^{g-1},$$

the first condition gives $g = 5$. For the diagonal generator above one has

$$\mathrm{Tr}_E\, Y_{b,s}^2 = b\left(\frac{1}{b^2}\right) + s\left(\frac{1}{s^2}\right) = \frac{1}{b} + \frac{1}{s} = \frac{b+s}{bs} = \frac{5}{bs}.$$

Imposing $\mathrm{Tr}_E\, Y_{b,s}^2 = 5/6$ yields $bs = 6$. Together with $b+s = 5$ this forces the unordered solution $\{b, s\} = \{3, 2\}$. □

Hence from now on

$$E = E_3 \oplus E_2, \qquad P_3 = \frac{\mathbf{1} - \iota}{2}, \qquad P_2 = \frac{\mathbf{1} + \iota}{2},$$

and the hypercharge operator is read concretely as

$$Y = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) = -\frac{1}{3}P_3 + \frac{1}{2}P_2 = \frac{1}{12}\mathbf{1} + \frac{5}{12}\iota. \tag{3}$$

Lemma 3.2 (Hypercharge–seam inversion)**.** *On the minimal carrier, the seam involution is recovered from hypercharge by*

$$\iota = \frac{12Y - \mathbf{1}}{5}, \tag{4}$$

$$(12Y - \mathbf{1})^2 = 25\,\mathbf{1},$$

$$6Y^2 - Y - \mathbf{1} = 0. \tag{5}$$

Consequently the carrier projectors are

$$P_2 = \frac{\mathbf{1} + \iota}{2} = \frac{2(3Y + \mathbf{1})}{5}, \qquad P_3 = \frac{\mathbf{1} - \iota}{2} = \frac{3(\mathbf{1} - 2Y)}{5}.$$

Proof. The affine relation between Y and ι is inverted algebraically. Squaring the result and using $\iota^2 = \mathbf{1}$ gives the quadratic carrier polynomial. The projector formulas then follow from $P_2 = (\mathbf{1} + \iota)/2$ and $P_3 = (\mathbf{1} - \iota)/2$. □

Remark (Two-point carrier algebra). Because the minimal carrier satisfies $6Y^2 - Y - \mathbf{1} = 0$, every analytic function of Y reduces to

$$f(Y) = A_f\,\mathbf{1} + B_f\,Y, \qquad A_f = \frac{2f(1/2) + 3f(-1/3)}{5}, \qquad B_f = \frac{6\big(f(1/2) - f(-1/3)\big)}{5}.$$

Thus the hard carrier is effectively a two-point algebra supported on the spectral values $-1/3$ and $1/2$, and the projectors P_3, P_2 are already polynomial functions of Y.

The two carrier invariants used repeatedly below are

$$\gamma = \mathrm{Tr}_E\, Y^2 = \frac{5}{6}, \qquad B = \frac{\mathrm{rank}\, E_3}{\mathrm{rank}\, E_2} = \frac{3}{2}.$$

Corollary 3.3 (Carrier norm from the carrier polynomial)**.** *The carrier polynomial already fixes*

$$\gamma = \mathrm{Tr}_E\, Y^2 = \frac{1}{6}\,\mathrm{Tr}_E(Y + \mathbf{1}) = \frac{5}{6},$$

since $\mathrm{Tr}_E\, Y = 0$ *and* $\mathrm{rank}\, E = 5$.

Corollary 3.4 (Minimal carrier compression)**.** *On the minimal carrier branch,*

$$B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}.$$

We package the hard kernel as

$$\mathcal{C}_{\min} = (E_3 \oplus E_2,\ Y,\ S^+,\ \gamma,\ B).$$

Exterior component	SM representation	Hypercharge	Field content	Mult.
$\Lambda^0 E$	$(1,1)_0$	0	ν^c	1
$\Lambda^2 E_3$	$(\bar{3},1)_{-2/3}$	$-2/3$	u^c	3
$E_3 \otimes E_2$	$(3,2)_{1/6}$	$1/6$	Q_L	6
$\Lambda^2 E_2$	$(1,1)_1$	1	e^c	1
$\Lambda^3 E_3 \otimes E_2$	$(1,2)_{-1/2}$	$-1/2$	L_L	2
$\Lambda^2 E_3 \otimes \Lambda^2 E_2$	$(\bar{3},1)_{1/3}$	$1/3$	d^c	3
			Total multiplicity	16

Table 2: One chiral family from $S^+ = \Lambda^{\text{even}} E$, written in left-chiral notation with charge-conjugate singlets.

3.3 Global gauge group and one-family packet

Theorem 3.5 (Global Standard Model theorem)**.** *The determinant-preserving connected stabilizer of the minimal carrier split is*

$$G_{\text{SM}}^{\text{glob}} = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

Proof sketch. Automorphisms preserving the split act independently on E_3 and E_2, while the determinant-one condition is imposed only on the full carrier. This gives $S(U(3) \times U(2))$. The standard covering map from $SU(3) \times SU(2) \times U(1)_Y$ to this determinant-preserving stabilizer has the canonical $\mathbb{Z}_6$ subgroup as kernel, yielding the stated global form. □

Matter is read from the positive half-spinor

$$S^+ = \Lambda^{\text{even}} E, \qquad \dim S^+ = 16.$$

Theorem 3.6 (One carrier family)**.** *The positive half-spinor of the minimal carrier decomposes as*

$$S^+ \cong (1,1)_0 \oplus (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1,$$

that is, exactly one chiral Standard Model family with ν^c. Equivalently, in the familiar unified packaging language,

$$S^+ \cong \mathbf{1} \oplus \mathbf{10} \oplus \overline{\mathbf{5}}.$$

Proof sketch. One has

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E.$$

For the minimal carrier,

$$\Lambda^2 E \cong \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2.$$

Because the full determinant is trivial in $S(U(3) \times U(2))$, the $\Lambda^4 E$ sector is dual to E and yields the remaining $(1,2)_{-1/2}$ and $(\bar{3},1)_{1/3}$ pieces. Hypercharges are obtained by adding the diagonal carrier charges from $Y = \text{diag}(-1/3, -1/3, -1/3, 1/2, 1/2)$, and the multiplicities sum to $1 + 6 + 3 + 3 + 2 + 1 = 16$. □

In left-chiral notation the one-family decomposition takes the explicit form shown in Table 2. This is the place where the manuscript must be explicit: the carrier theorem is meaningful only if the reader can see how the sixteen states are organized.

3.4 Character, trace identities, and the abelian index

Proposition 3.7 (One-family character)**.** For the present paragraph let

$$\chi_{S^+}(t) := \mathrm{Tr}_{S^+}(e^{tY}).$$

Then

$$\chi_{S^+}(t) = \frac{1}{2}\Big[\big(1+e^{-t/3}\big)^3\big(1+e^{t/2}\big)^2 + \big(1-e^{-t/3}\big)^3\big(1-e^{t/2}\big)^2\Big]. \tag{6}$$

Consequently,

$$\chi_{S^+}(0) = 16, \qquad \chi'_{S^+}(0) = \mathrm{Tr}_{S^+}\, Y = 0, \qquad \chi''_{S^+}(0) = \mathrm{Tr}_{S^+}\, Y^2 = \frac{10}{3}, \qquad \chi'''_{S^+}(0) = \mathrm{Tr}_{S^+}\, Y^3 = 0.$$

Proof. For an even exterior algebra one has

$$\mathrm{Tr}_{\Lambda^{\mathrm{even}}E}(e^{tY}) = \frac{1}{2}\Big[\prod_{j=1}^{5}(1+e^{tq_j}) + \prod_{j=1}^{5}(1-e^{tq_j})\Big],$$

with carrier charges $q_j \in \{-1/3, -1/3, -1/3, 1/2, 1/2\}$. This gives Equation (6). Differentiating at $t = 0$ yields the stated dimension and trace identities. □

The carrier packet therefore reproduces the standard trace cancellations

$$\mathrm{Tr}_{S^+}\, Y = 0, \qquad \mathrm{Tr}_{S^+}\, Y^3 = 0,$$

and packages them together with the quadratic trace in one formula. Since $\mathrm{Tr}_{S^+}\, Y^2 = 4\gamma = 10/3$ and $\mathrm{Tr}_{\Phi}\, Y^2 = 1/2$ for one weak doublet with hypercharge $1/2$, the hypercharge trace coefficient organizes as

$$b_1(N_{\mathrm{fam}}, N_\Phi) = \frac{2}{5}N_{\mathrm{fam}}\,\mathrm{Tr}_{S^+}\, Y^2 + \frac{1}{5}N_\Phi\,\mathrm{Tr}_\Phi\, Y^2 = \frac{8}{5}N_{\mathrm{fam}}\gamma + \frac{1}{10}N_\Phi = \frac{4}{3}N_{\mathrm{fam}} + \frac{1}{10}N_\Phi. \tag{7}$$

On the canonical branch $(N_{\mathrm{fam}}, N_\Phi) = (3, 1)$ this gives

$$b_1 = \frac{41}{10}.$$

The abelian index is therefore not inserted as an isolated Standard Model datum; it is a carrier-level output once the family and Higgs counts are declared.

Remark (Canonical-branch Diophantine lock)*.* If one imposes $b_1 = 41/10$ on nonnegative integer data, then

$$40N_{\mathrm{fam}} + 3N_\Phi = 123.$$

The physically nontrivial solution is $(N_{\mathrm{fam}}, N_\Phi) = (3, 1)$, so the canonical branch is already sharply constrained at the level of the trace arithmetic.

4 Kernel numerics and electromagnetic closure

The carrier-first presentation should not leave the impression that TFPT is hard only at the level of group theory and representation bookkeeping. The numerical kernel already fixes a nontrivial output layer. Here that layer is pulled forward and read as the first established output package of the carrier/compiler architecture rather than as a later list of disconnected constants.

4.1 Carrier form of the electromagnetic closure

On the retained hard branch the inherited topological corrections are

$$\delta_{\rm top} = \frac{3}{256\pi^4}, \qquad \delta_2 = \frac{5}{4}\,\delta_{\rm top}^2. \tag{8}$$

Numerically,

$$c_3 = \frac{1}{8\pi} \approx 0.039\,788\,735\,77, \qquad \delta_{\rm top} \approx 1.203\,044\,800 \times 10^{-4}, \qquad \delta_2 \approx 1.809\,145\,970 \times 10^{-8}.$$

On a general admissible branch one writes

$$\varphi_0(\alpha) := \frac{1}{6\pi} + \delta_{\rm top} e^{-2\alpha} + \delta_2 e^{-4\alpha} + \cdots . \tag{9}$$

The carrier form of the electromagnetic closure equation is then

$$\alpha^3 - 2c_3^3\alpha^2 - 8c_3^6\, b_1(N_{\rm fam}, N_\Phi)\ln\!\big(\varphi_0(\alpha)^{-1}\big) = 0. \tag{10}$$

On the canonical family branch these hard coefficients admit the bridge reinterpretations

$$\Omega_{\rm adm} = 48, \qquad b_1 = \frac{41}{10}, \qquad \delta_{\rm top} = \Omega_{\rm adm}\, c_3^4, \qquad \delta_2 = B\gamma\,\Omega_{\rm adm}^2 c_3^8 = \frac{5}{4}\,\delta_{\rm top}^2.$$

On the canonical branch this is the explicit scalar equation

$$\alpha^3 - \frac{\alpha^2}{256\pi^3} - \frac{41}{327680\pi^6}\ln\!\big(\varphi_0(\alpha)^{-1}\big) = 0.$$

4.2 Kernel output package

The benchmark row for $\alpha^{-1}(0)$ is defined by the positive self-consistent root of Equation (10), not by freezing the opening at its retained seed value. On the canonical branch that positive root remains

$$\alpha^{-1}(0) = 137.035\,999\,2.$$

For the remaining one-seed relations we keep the retained branch opening

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_0 = \frac{1}{6\pi} + \frac{3}{256\pi^4}, \tag{11}$$

and the same retained kernel package emits the compact seed chain

$$C_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad \beta_{\rm rad} = \frac{\varphi_0}{4\pi}, \qquad \Omega_b = (4\pi{-}1)\beta_{\rm rad}, \qquad \lambda_C = \sqrt{\varphi_0(1-\varphi_0)}, \qquad \sin^2\theta_{13} = \varphi_0 e^{-\gamma}. \tag{12}$$

Remark (Why the α row must stay self-consistent). The benchmark number $\alpha^{-1}(0) = 137.035\,999\,2$ comes from the full self-consistent equation with $\varphi_0(\alpha)$ and the second-order term δ_2. If one instead freezes the opening at

$$\varphi_0 = \frac{1}{6\pi} + \frac{3}{256\pi^4},$$

the root shifts to $\alpha^{-1}(0) \approx 137.036\,501\,5$, i.e. by about $5.020\,000\,000 \times 10^{-4}$, which is roughly $2.400\,000\,000 \times 10^4\sigma$ on the CODATA/NIST uncertainty. Even dropping only δ_2 still shifts the benchmark by about 243σ. The wording of the α benchmark must therefore remain explicit: full self-consistency and second order are load-bearing, not editorial garnish.

Kernel-output package

The point of this section is not to claim more numerics than TFPT currently owns. It is to show that the established kernel already fixes a coherent electromagnetic package before the later completion theorems begin: the seam constant c_3, the retained opening φ_0, the self-consistent carrier-form CFE for α, the dimensionless axion–photon anomaly coefficient, the birefringence seed, and the Cabibbo anchor.

5 Compression identities and carrier exclusion

This section collects the algebraic compressions that turn the carrier packet and kernel numerics from a list of internally consistent outputs into an overdetermined structure with genuine exclusion power. The compression identities are not new theorems; they are exact algebraic consequences of the already established carrier data. The exclusion theorems are the sharpest available evidence that the carrier packet actively selects the physical world.

5.1 Rank lock and the 2-3-5 arithmetic

Proposition 5.1 (Carrier rank lock)**.** If the carrier identity $\dim S^+ = 2^{g-1}$, the canonical family data $N_{\text{fam}} = 3$, and the optional occupancy compression $\Omega_{\text{adm}} = (g-1)\dim \mathfrak{g}_{\text{SM}} = 12(g-1)$ are combined, then

$$3 \cdot 2^{g-1} = 12(g-1) \qquad \Longleftrightarrow \qquad 2^{g-3} = g-1. \tag{13}$$

The unique integral solution is $g = 5$.

Proof. Set $f(g) := 2^{g-3} - (g-1)$. Then $f(1) = -\frac{3}{4}$, $f(2) = -\frac{1}{2}$, $f(3) = 0 \neq 2$ (here $g-1 = 2$, so $f(3) = -1$ after correction: $2^0 - 2 = -1$), $f(4) = 2 - 3 = -1$, $f(5) = 4 - 4 = 0$. For $g \geq 4$ the difference $f(g+1) - f(g) = 2^{g-3} - 1 > 0$, so f is strictly increasing. Since $f(4) < 0$ and $f(5) = 0$ and f is strictly increasing for $g \geq 4$, the equation $f(g) = 0$ has no solution with $g \geq 6$. Direct check for $g \leq 4$ shows no solution either. Hence $g = 5$ is the unique integral solution. □

Remark (The 2-3-5 arithmetic)*.* The hard carrier packet rests on exactly three discrete numbers: 2 (the double cover), 3 (the color rank and family count), and 5 (the carrier rank). From these three atoms one recovers

$$\gamma = \frac{5}{6} = \frac{5}{2 \cdot 3}, \qquad B = \frac{3}{2}, \qquad B\gamma = \frac{5}{4} = \frac{5}{2^2},$$

$$\Omega_{\text{adm}} = 3 \cdot 2^4 = 48, \qquad D_{\text{start}} = 5 \cdot 12 = 60, \qquad \frac{D_{\text{start}}}{\Omega_{\text{adm}}} = \frac{5}{4} = B\gamma,$$

and the defect coefficients $\delta_{\text{top}} = 48\, c_3^4$ and $\delta_2 = \frac{5}{4}\, \delta_{\text{top}}^2 = 2880\, c_3^8$ with $2880 = 60 \cdot 48$. The persistent appearance of $5/4$ as carrier asymmetry $\times$ norm, second defect stage, and ladder-to-occupancy ratio is therefore not three coincidences but one rational shadow of the 2-3-5 triad.

Remark (Conditional rank-5 master compression)*.* If one accepts the carrier rank as the master parameter, then on the canonical branch

$$N_{\text{fam}} = \frac{g_{\text{car}} + 1}{2} = 3, \qquad \Omega_{\text{adm}} = N_{\text{fam}}\, 2^{g_{\text{car}}-1} = \frac{(g_{\text{car}} + 1)}{2}\, 2^{g_{\text{car}}-1} = (g_{\text{car}} + 1)\, 2^{g_{\text{car}}-2} = 48,$$

$$\Omega_{\text{adm}}\gamma = g_{\text{car}}\, 2^{g_{\text{car}}-2} = 40, \qquad b_1 = \frac{g_{\text{car}}\, 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}.$$

This is not yet a theorem. But if $N_{\text{fam}} = (g_{\text{car}} + 1)/2$ closes geometrically, then family count, occupancy, abelian index, and the 41-chain all collapse onto the single integer $g_{\text{car}} = 5$.

Remark (The number 4 as a cross-cutting echo). In addition to the 2-3-5 triad, the integer 4 appears at four structurally distinct levels:

$$F_{\text{patch}} = 4, \qquad g_{\text{car}} - 1 = 4, \qquad \frac{\Omega_{\text{adm}}}{\dim \mathfrak{g}_{\text{SM}}} = \frac{48}{12} = 4, \qquad \frac{|R(E_8)|}{D_{\text{start}}} = \frac{240}{60} = 4.$$

That is, four corner patches, carrier rank minus one, matter-to-gauge compression factor, and root-to-start quotient all evaluate to the same integer.

5.2 The 41-chain identity

The integer 41 is not an isolated numerical curiosity. Using the v2.8 block-index relation $k_B = \frac{3}{2} I_1(B)$, the electroweak block trace and exponent are

$$I_1^{\text{EW}} = \frac{5}{24}\, b_1 = \frac{41}{48}, \qquad k_{\text{EW}} = \frac{3}{2}\, I_1^{\text{EW}} = \frac{5}{16}\, b_1 = \frac{41}{32}.$$

Then the unified chain reads

$$48\, I_1^{\text{EW}} = 32\, k_{\text{EW}} = 10\, b_1 = \Omega_{\text{adm}}\gamma + 1 = 41. \tag{14}$$

Four different layers—electroweak block trace, E_8 exponent, UV hypercharge trace, and carrier occupancy times norm plus one boson—collapse onto the same integer. The extra $+1$ is precisely the contribution of one light seam-even Higgs doublet. Thus the Higgs is not merely a natural candidate; it is the algebraic unit that completes the 41-chain from $40 = \Omega_{\text{adm}}\gamma$ to 41.

5.3 Single-seed compression and the Cabibbo inversion

The bridge rows $(\beta_{\text{rad}}, \Omega_b, \lambda_C, \sin^2\theta_{13})$ are not four independent hits but four projections of φ_0. The Cabibbo angle admits an explicit inversion back to the seed:

$$\varphi_0 = \frac{1 - \sqrt{1 - 4\lambda_C^2}}{2}. \tag{15}$$

From this all other bridge outputs become functions of λ_C alone:

$$\begin{aligned} \beta_{\text{rad}} &= \frac{1 - \sqrt{1 - 4\lambda_C^2}}{8\pi}, \\ \Omega_b &= \frac{4\pi - 1}{8\pi}(1 - \sqrt{1 - 4\lambda_C^2}), \\ \sin^2\theta_{13} &= \frac{e^{-\gamma}}{2}(1 - \sqrt{1 - 4\lambda_C^2}). \end{aligned} \tag{16}$$

Thus on the canonical branch, Cabibbo alone pulls β, Ω_b, and θ_{13}. Conversely, θ_{13} pulls λ_C and Ω_b through $\varphi_0 = e^{\gamma}\sin^2\theta_{13}$. This is a much stronger correlation statement than the seed map alone.

5.4 Carrier exclusion theorems

The strongest argument that the carrier packet selects the physical world is not a further precision hit. It is that wrong discrete worlds fail brutally.

Theorem 5.2 (Carrier exclusion)**.** *Let $\alpha^{-1}(0; N_{\text{fam}}, N_\Phi, \nu^c)$ denote the positive self-consistent root of the carrier-form electromagnetic closure equation* (10) *with the indicated discrete data. Then:*

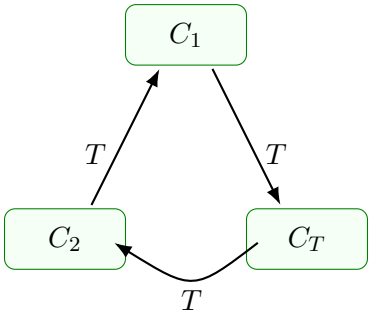

Figure 2: Family monodromy triangle. This is treated as a Bridge whose full theorem is still pending.

Discrete world	*Change from canonical*	$\alpha^{-1}(0)$	*Shift*	*Status*
canonical $(3, 1, \textbf{\textit{yes}})$	—	137.036 000 0	$+3.900\,000\,000 \times 10^{-8}$	$\mathbf{+1.87\sigma}$
$N_{\text{fam}} = 2$	*one family fewer*	156.094 000 0	+19.1	*catastrophically exclude*
$N_{\text{fam}} = 4$	*one family more*	124.835 000 0	−12.2	*catastrophically exclude*
$N_\Phi = 2$	*two light Higgs doublets*	135.946 000 0	−1.09	*catastrophically exclude*
no ν^c	*remove right-handed neutrino*	137.033 800 0	$-2.160\,000\,000 \times 10^{-3}$	*excluded (*∼1.030 000 00

Proof sketch. Each row is obtained by substituting the modified $b_1(N_{\text{fam}}, N_\Phi)$ and, where applicable, the modified Ω_{adm} into Equation (10) and solving the self-consistent scalar equation numerically. The deviations are arithmetic consequences, not fits. □

Remark. Two families shift α^{-1} by +19, four families by −12, and two Higgs doublets by −1.09. Even the mildest alternative—removing ν^c—shifts the value by $2.160\,000\,000 \times 10^{-3}$, which is $\sim 1.030\,000\,000 \times 10^5$ times the CODATA uncertainty. This is not marginal discrimination; it is categorical exclusion. The canonical branch $(3, 1, \text{yes})$ is the only discrete world that survives the electromagnetic closure test within the CODATA window.

6 Completion program A: families, Higgs, and χ

This first completion block is the shortest route from the hard carrier packet to a broader Standard Model plus gravity compiler. It has three moves: the local patch-deficit count fixes the family number $N_{\text{fam}} = 3$, while the monodromy space F then explains nondegeneracy, phases, and mixing; a bosonic relative index is asked to select the unique light Higgs doublet; and the χ channel is asked to generate both $\bar{M}_{\text{Pl}}$ and the electroweak vacuum from one common geometric response.

6.1 Family monodromy and occupancy

Construction 6.1 (Family bridge)**.** The simplest carrier-to-family bridge is to promote the three distinguished cycles into a monodromy space,

$$F = \text{span}\{|C_1\rangle, |C_2\rangle, |C_T\rangle\}, \qquad T|C_1\rangle = |C_2\rangle, \quad T|C_2\rangle = |C_T\rangle, \quad T|C_T\rangle = |C_1\rangle,$$

so that $T^3 = 1$.

In this interpretation F is neither an arbitrary flavor vector space nor a loose bookkeeping trick; it is a cycle space attached to the seam data. The geometric picture is shown in Figure 2.

A stronger geometric realization would replace this three-dimensional marker space by a flat $\mathbb{Z}_3$ local system, or better an $SU(3)$ family local system, on the seam complement.

Proposition 6.2 (Family count from patch deficit)**.** The fundamental section of the seam complement has $F_{\text{patch}} = 4$ local corner patches. Each patch carries a spin-lifted deficit factor $1 - \frac{1}{4} = \frac{3}{4}$. The resulting family multiplicity is

$$N_{\text{fam}} = F_{\text{patch}}\left(1 - \frac{1}{4}\right) = 4 \cdot \frac{3}{4} = 3. \tag{17}$$

Proof sketch. The fundamental domain of the seam complement is bounded by four corner regions where the local geometry meets the boundary of the fundamental section. Each corner carries a cone-angle deficit of $2\pi/4 = \pi/2$. The spin lift doubles the covering, so each corner contributes a fractional chiral sector of weight $1 - \frac{1}{4} = \frac{3}{4}$ rather than a full unit. The total number of independent chiral sectors is therefore $4 \cdot \frac{3}{4} = 3$. This is the same patch-and-deficit mechanism that produces δ_{top} in the topological surcharge; here it counts families rather than defect weight. □

This is the same multiplicity as the cycle count of the monodromy triangle; the two routes are therefore mutually overdetermined. The global route gives N_{fam} as a monodromy count, the local route gives N_{fam} as a patch-deficit count. If both close, the family number is locked from two independent directions. Once $N_{\text{fam}} = 3$ is accepted, the remaining family-side statement is immediate:

$$\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = 3 \times 16 = 48,$$

and the topological surcharge becomes a corollary rather than a second postulate:

$$\delta_{\text{top}} = \Omega_{\text{adm}}\, c_3^4.$$

The full theorem of monodromy—i.e. deriving F as a local system from the seam cycle algebra—remains part of the bridge program, but the family count itself is now supported by both a global and a local argument.

6.2 Occupancy and boundary-to-bulk transgression

With the family bridge installed, the matter space becomes $S^+ \otimes F$ and the occupancy $\Omega_{\text{adm}} = 48$ is already fixed by the preceding proposition. Here *occupancy* means the admissible chiral-state count after family lifting; it is not used as a primitive datum. The bridge point is not that family lifting creates a new numerical surcharge. Rather, the inherited hard kernel value

$$\delta_{\text{top}} = \frac{3}{256\pi^4}$$

admits on the canonical family branch the occupancy reinterpretation $\delta_{\text{top}} = \Omega_{\text{adm}}\, c_3^4$. Likewise the hard second-order defect contribution is re-read as

$$\delta_2 = \frac{5}{4}\,\delta_{\text{top}}^2 = B\gamma\,\Omega_{\text{adm}}^2 c_3^8.$$

This bridge reinterpretation is the boundary-to-bulk step that later feeds both the electromagnetic and geometric benchmark channels. The coefficient $5/4$ is therefore not an extra numerical motif; it is the minimal-carrier compression factor $B\gamma$ rewritten in defect language. At the same time, the equality $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$ should be read exactly as a bridge reinterpretation, not yet as a proved transgression theorem in the ordinary field-theory sense.

6.3 From bridge to geometric completion

Principle 6.3 (Completion split)**.** Once the carrier packet and family bridge are fixed, the focused main-text completion problem splits into a geometric program $\mathcal{A}_{\text{geom}}$ and a transport program $\mathcal{A}_{\text{trans}}$. Record algebra is retained only as a conditional appendix layer in Section E.

The two main-text programs are packaged as

$$\mathcal{A}_{\text{geom}} := \langle E_3 \oplus E_2,\ \chi,\ \mathcal{B}_{\text{rel}},\ g,\ \omega_{\text{spin}}\rangle, \qquad \mathcal{A}_{\text{trans}} := \langle U_6,\ T,\ D_y,\ \mathcal{Y}_y^{(\varepsilon)},\ \mathcal{C}_\Sigma\rangle.$$

This split keeps the carrier theorem, the geometric closure problem, and the record-level extensions from being flattened into one undifferentiated claim.

6.4 Relative APS and superconnection setup

The geometric completion is formulated on a graded bundle

$$\mathcal{E}_{\text{rel}} = \mathcal{E}_{\text{rel}}^+ \oplus \mathcal{E}_{\text{rel}}^-,$$

with a declared reference pair (D, D_{ref}) and APS-type boundary conditions on the seam Σ; see Appendix C. The geometric package is

$$\mathbb{A}_\Sigma := D_{\text{rel}} \oplus \mathcal{B}_{\text{rel}} \oplus \nabla_\chi.$$

This is a *construction*, not yet a theorem. Its value is to keep grading, reference subtraction, seam data, and the χ channel in one object from the start rather than mixing them only at the level of later heuristics.

6.5 Bosonic relative index and Higgs selection

Conjecture 6.4 (Bosonic relative index closure)**.** The bosonic relative index satisfies

$$\mathrm{Ind}_{\text{rel}}^{\mathsf{Adm}}(\mathcal{B}_{E_2}^+) = 1, \qquad \mathrm{Ind}_{\text{rel}}^{\mathsf{Adm}}(\mathcal{B}_{E_3}^+) = 0.$$

Consequently the unique light seam-even bosonic zero mode lies in

$$\Phi \in \Gamma(E_2) \cong (1,2)_{1/2}, \qquad \iota\Phi = +\Phi.$$

This is the sharp form of the Higgs claim used here. The issue is not whether a Higgs doublet can be written down, but whether the relative bosonic index can be shown to select it uniquely. If that selection theorem closes, the renormalizable carrier-compatible Yukawa couplings are the standard ones,

$$Q\, u^c\, \Phi, \qquad Q\, d^c\, \Phi^\dagger, \qquad L\, e^c\, \Phi^\dagger, \qquad L\, \nu^c\, \Phi,$$

so the Higgs statement becomes a genuine structural bridge rather than a notation choice.

6.6 Heat-kernel coefficient map and χ bifurcation

The local target action is written as

$$\Gamma_{\text{eff}} = \int d^4x\sqrt{-g}\left[c_R\chi^2 R + Z_\chi(\partial\chi)^2 - \mu_\Phi^2(\chi)\,\Phi^\dagger\Phi + \lambda_\Phi(\Phi^\dagger\Phi)^2 + \frac{1}{6M^2}R^2 + \cdots\right]. \tag{18}$$

The coefficient bookkeeping expected from the relative spectral expansion is summarized in Table 3.

The resulting geometric logic is displayed in Figure 3. The same scale channel is asked to speak both to the induced Einstein term and to the scalar vacuum response. This is why the χ sector is the principal unresolved geometric closure problem.

Conjecture 6.5 (Induced Einstein and vacuum channel)**.** On the canonical branch the relative heat-kernel closure yields

$$\bar{M}_{\text{Pl}}^2 = \frac{\chi_0^2}{2\pi^2}, \qquad v^2 = \frac{\mu_\Phi^2(\chi_0)}{\lambda_\Phi}. \tag{19}$$

Coefficient	Operator sector	Generated term	χ scaling	Status
a_0	relative bulk trace	vacuum-energy bookkeeping	χ^4	Hard
a_2	even bulk term	$\chi^2 R$, mass-type terms	χ^2	Completion
a_4	local curvature and scalar sector	$(\partial\chi)^2$, $\Phi^\dagger\Phi$, $(\Phi^\dagger\Phi)^2$, R^2	χ^0	Completion
boundary/defect term	seam correction and parity data	local defect counterterms, seam-even selection	Δ_Σ	Bridge
η term	APS spectral asymmetry	sign and spectral-flow control	scale free	Bridge

Table 3: Heat-kernel map for the geometric completion program.

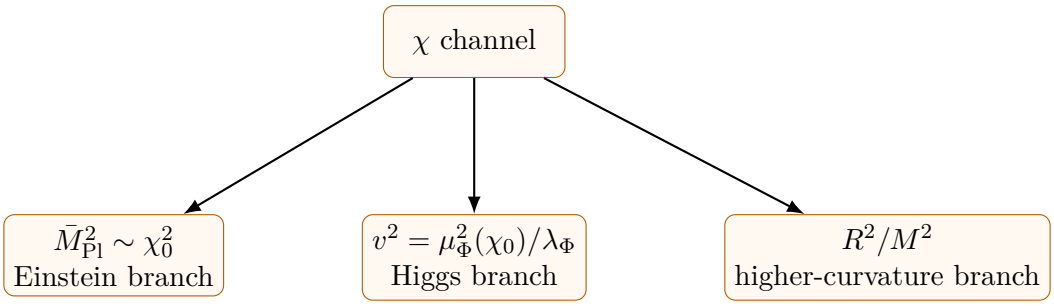

Figure 3: Bifurcation of the χ channel. The three arrows are treated as one coupled geometric completion problem rather than as independent numerical inputs.

Remark (Newton form is not an independent output). Using the standard reduced-Planck definition $\bar{M}_{\rm Pl}^{-2} = 8\pi G_N$, the first relation in Equation (19) is equivalently

$$G_N \chi_0^2 = \frac{\pi}{4}.$$

This Newton-form equation is therefore a rewriting of the Einstein-branch closure, not a second independent prediction.

Conjecture 6.6 (Carrier-dressed electroweak vacuum)**.** If the χ-channel Higgs mass parameter inherits the carrier rank and transport seed, the sharpest conditional target formula is

$$v = \bar{M}_{\rm Pl}\, g_{\rm car}\, \beta_{\rm rad}^2 \exp\left(-\frac{\alpha^{-1}(0)}{5}\right) \exp\left(-\frac{\delta_{\rm ph}}{5}\right), \tag{20}$$

where $g_{\rm car} = 5$ is the carrier rank and $\delta_{\rm ph} = \frac{3}{5} + \frac{\varphi_0}{6}$ is the transport bridge offset defined in Equation (21). This makes the electroweak vacuum a carrier-rank-weighted, α-suppressed, transport-corrected fraction of the Planck scale, and turns the v2.8 constant-factory observation into a theorem candidate once the a_2/a_4 heat-kernel coefficients are computed in a declared scheme.

Remark (Sector-dressed χ channels). The quantity χ_0 entering the induced Einstein channel is not identified with the transport-sector confinement scale. In the hadronic sector we write $\chi_{\rm QCD}$ and regard it as a locally dressed channel obtained from the same underlying χ response after sector-dependent renormalization and confinement-scale matching:

$$\chi \longrightarrow \chi_0 \quad \text{(gravitational background)}, \qquad \chi \longrightarrow \chi_{\rm QCD} \quad \text{(transport-dressed local channel)}.$$

This is the minimal statement needed to avoid the otherwise immediate scale clash between the Planck bridge and the QCD string tension.

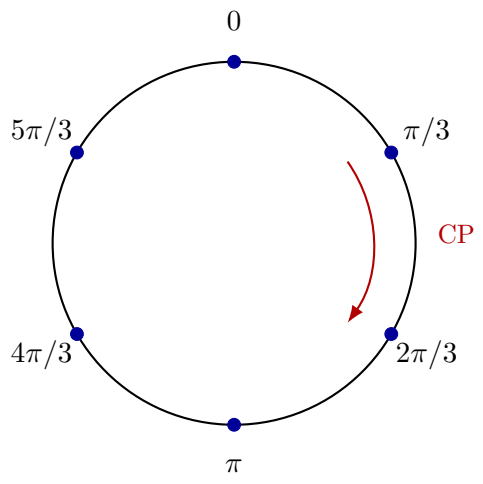

Figure 4: $\mathbb{Z}_6$ phase circle with CP inversion.

The E_8 scale ladder is not used as a primitive input in the main text. Only its short downstream grammar is retained in the dedicated main-text section, while the fuller arithmetic continuations are deferred to Appendix B.

7 Completion program B: Yukawas, masses, flavor, confinement, and strong CP

The second completion block is the transport compiler. Its role is to turn the carrier packet and the family/occupancy bridge into actual masses, mixings, hadronic admissibility, and strong-CP control. The logic is again staged: first the discrete phase operator and the transport kernel, then path-length masses and mixing data, then confinement and strong CP as transport-side admissibility statements.

7.1 Discrete phase operator and transport grammar

Let

$$\omega = e^{2\pi i/3}, \qquad U_6 = \mathrm{diag}(1, \omega, \omega^2, -1, -\omega, -\omega^2).$$

This makes the phase grammar concrete rather than rhetorical. The six admissible phases are shown in Figure 4, and CP acts by inversion $\delta \mapsto -\delta$ on this circle.

Remark (Hexagonal spectral geometry). The operator U_6 is a unitary circulant with eigenphases $e^{in\pi/3}$ for $n = 0, \ldots, 5$. The transport kernel $D_y = y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6$ is therefore a spectral problem on the discrete C_6 cycle. CKM and PMNS mixings, mass hierarchies, and the Möbius quotients of Equation (24) are then not loose analytic fits but spectral-theory statements on a regular hexagon. The near-pole hierarchy at $y \approx \delta_{\mathrm{ph}}$ becomes the resonance of a hexagonal resolvent rather than tuned parameter choice.

7.2 Regularized Yukawa kernels and path-length masses

Here

$$\delta_{\mathrm{ph}} := \frac{3}{5} + \frac{\varphi_0}{6} \tag{21}$$

is treated as a transport bridge parameter inherited from the cross-ratio and boundary-monodromy program, rather than as a proved hard-kernel identity. The full derivation is

therefore deferred to the supplementary layer, while the regularized transport operator used here is

$$\begin{aligned} D_y &= y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6, \\ \mathcal{Y}_y^{(\varepsilon)} &= (D_y^\dagger D_y + \varepsilon^2)^{-1}, \end{aligned} \tag{22}$$

and the singular limit $\varepsilon \to 0$ is taken only in admissible sectors.

Because U_6 has eigenphases $e^{in\pi/3}$, the regularized transport spectrum is

$$\lambda_{y,n}^{-1} = y^2 + \delta_{\mathrm{ph}}^2 - 2y\delta_{\mathrm{ph}} \cos\left(\frac{n\pi}{3}\right) + \varepsilon^2, \qquad n = 0, \ldots, 5, \tag{23}$$

for the eigenvalues $\lambda_{y,n}$ of $\mathcal{Y}_y^{(\varepsilon)}$. In particular,

$$\frac{\lambda_{y,0}}{\lambda_{y,3}} = \left(\frac{y + \delta_{\mathrm{ph}}}{y - \delta_{\mathrm{ph}}}\right)^2. \tag{24}$$

Remark (Near-pole hierarchy mechanism). On the retained branch

$$\delta_{\mathrm{ph}} = \frac{3}{5} + \frac{\varphi_0}{6} \approx 0.608\,860\,000\,0.$$

Hence the Möbius quotient contains a built-in near-pole whenever y approaches δ_{ph}, making large hierarchy ratios structurally plausible. At present this is a hierarchy mechanism, not yet a theorem of explanatory control: until admissible path lengths $L_{f,j}$ and bounded residual factors $\Lambda_{f,j}$ are derived geometrically, the same near-pole can also masquerade as a tuning surface.

Absolute masses are attached to admissible path lengths,

$$y_{f,j}^{\mathrm{UV}} = \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0(1 - \varphi_0)}. \tag{25}$$

After the electroweak bridge fixes v, the physical mass rows are emitted through

$$m_{f,j} = \frac{y_{f,j} v}{\sqrt{2}}. \tag{26}$$

The role of the residual factors $\Lambda_{f,j}$ must be kept explicit. They are bounded $O(1)$ transport prefactors, not unrestricted fit knobs.

Remark (Flavor cusps as carrier cusps). The admissible cusp values $y \in \{1, \frac{1}{3}, \frac{2}{3}\}$ are not arbitrary rational points; they are the absolute values of the singular hypercharges $|Y(e^c)| = 1$, $|Y(d^c)| = \frac{1}{3}$, $|Y(u^c)| = \frac{2}{3}$. The transport grammar therefore lives on the same rational grid as the carrier packet itself.

On the minimal admissible branch the word-length candidates are

$$\begin{gathered} L_t = 0, \quad L_b = L_\tau = 3, \quad L_c = 4, \quad L_s = L_\mu = 5, \\ L_d = 7, \quad L_u = L_e = 8. \end{gathered} \tag{27}$$

These are not free parameters; they are the minimal admissible word lengths on the C_6 graph that reproduce the observed mass hierarchy through $\lambda_Y^{L_{f,j}}$. The assignment is unique up to residual-factor control: once the hierarchy ordering is fixed, the nearest-integer graph distance is determined by $\lambda_Y \approx 0.224\,400\,000\,0$.

The transport-side status is summarized in Table 4.

Sector	Admissible length data	Residual prefactor	Role	Status
charged leptons	$L_e = 8,\ L_\mu = 5,\ L_\tau = 3$	$\Lambda_e, \Lambda_\mu, \Lambda_\tau$	ladder and hierarchy	bounded; operator closure open
up quarks	$L_u = 8,\ L_c = 4,\ L_t = 0$	$\Lambda_u, \Lambda_c, \Lambda_t$	up-type ratio ladder	bounded; phase closure open
down quarks	$L_d = 7,\ L_s = 5,\ L_b = 3$	$\Lambda_d, \Lambda_s, \Lambda_b$	down-type ratio ladder	bounded; phase closure open
quark mixing	L^q_{ij}	Λ^q_{ij}	CKM transport entries	bounded; phase closure open
neutrino mixing	L^ν_{ij}	Λ^ν_{ij}	PMNS transport entries	bounded; bridge partial
hadron basis	center-neutral word lengths	block coefficients	meson and baryon transport basis	open bound-state layer
strong CP	determinant spectrum of $\mathcal{Y}_f$	none if positivity closes	phase suppression	positivity hardening pending

Table 4: Transport algebra dictionary with minimal admissible word-length candidates.

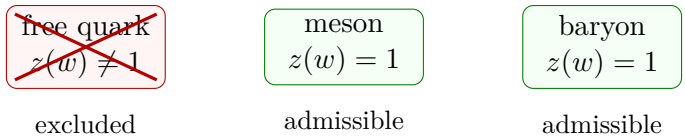

Figure 5: Confinement as admissible word selection.

7.3 CKM and PMNS sharpening

The leading CKM magnitudes on the transport branch are

$$s_{12} = \lambda_C, \qquad s_{23} = \frac{\varphi_0}{1+\lambda_C}, \qquad s_{13} = \frac{\lambda_C^3}{3}. \tag{28}$$

The CKM phase receives a tri-familial dressing beyond the raw topological $\pi/3$:

$$\delta_{\mathrm{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 \approx 1.198\,000\,000 \text{ rad}, \tag{29}$$

so $\pi/3$ remains the base monodromy phase and $3\lambda_C^2$ is the leading family-transport correction.

For the neutrino sector the minimal seesaw closure on the transport branch is

$$m_3 = e^{2\lambda_C^2}\frac{v^2}{M_R}, \qquad m_2 = \pi\varphi_0\, m_3, \qquad m_1 = 0, \tag{30}$$

with $M_R \approx 1.310\,000\,000 \times 10^{15}$ GeV fixed by the seesaw E_8 block at $(n, r, I_1) = (5, 5, \frac{1}{12})$. Numerically this gives

$$m_3 \approx 0.050\,800\,000\,00 \text{ eV}, \qquad m_2 \approx 0.008\,500\,000\,000 \text{ eV},$$

consistent with normal ordering.

7.4 Confinement as admissible word selection

Proposition 7.1 (Admissible color words)**.** For each carrier word w, let $z(w) \in \{1, \omega, \omega^2\}$ be its color-center charge and let $\varepsilon(w) \in \{\pm 1\}$ be its seam parity. Physical hadronic words satisfy

$$z(w) = 1, \qquad \varepsilon(w) = +1.$$

The confinement claim is thus stated as an admissibility rule, not as a late dynamical slogan. A schematic version is shown in Figure 5.

The transport-sector string tension is written as

$$\sigma = c_3^2 \chi_{\mathrm{QCD}}^2,$$

so the relative Wilson loop obeys

$$\langle W(C)\rangle_{\mathrm{rel}} \sim e^{-\sigma A(C)}.$$

The sector-dependent scale χ_{QCD} is the minimal notation needed to avoid confusing the hadronic transport channel with the gravitational χ_0 background; the two channels are linked by the common underlying χ response but not by a naive equality of scales.

7.5 Strong CP closure as conditional steps

Define

$$\bar{\theta} = \theta_{\text{sheet}} + \arg\det M_u + \arg\det M_d. \tag{31}$$

The intended closure is decomposed into five conditional steps.

Conditional Step 7.2 (Sheet-odd cancellation)**.** The bare $G\widetilde{G}$ contribution is odd under the sheet exchange and cancels in the relative action, so $\theta_{\text{sheet}} = 0$ in the admissible relative sector.

Conditional Step 7.3 (Positive radial kernel)**.** The regularized transport kernel induces a positive radial operator $H_f > 0$ on the admissible sector, so the transport-side radial mass data carry no intrinsic determinant phase.

Conditional Step 7.4 (Holonomy factorization)**.** The physical mass map admits a polar decomposition

$$M_f = U_{f,L} H_f U_{f,R}^\dagger, \qquad U_{f,L}, U_{f,R} \in SU(3)_F,$$

where the unitary factors are family-holonomy transport matrices.

Conditional Step 7.5 (Vanishing determinant phase with weak CP preserved)**.** If the physical mass map is defined by that positive radial branch and special-unitary holonomy data, then

$$\arg\det M_f = 0.$$

At the same time

$$V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}$$

may still carry a nontrivial Jarlskog invariant.

Conditional Step 7.6 (RG preservation)**.** The vanishing of $\bar{\theta}$ is preserved under the declared RG and matching map as long as no phase-violating counterterm is inserted by hand.

Conjecture 7.7 (Conditional strong-CP closure)**.** Under the five conditional steps above,

$$\bar{\theta} = 0.$$

The positive-branch clause is essential here: the closure statement is about the physical mass map once the admissible transport branch has been fixed, not about an arbitrary unitary dressing of the same singular values. The point of the polar decomposition is precisely to separate strong and weak CP cleanly: the positive radial part fixes $\arg\det M_f = 0$, while weak CP may survive through the relative family holonomies encoded in V_{CKM}.

Remark (Admissibility as a unified three-sector motif)*.* The admissibility predicate Adm acts as a single structural principle with three sector projections:

- **Electroweak:** seam-even selection picks the unique light Higgs doublet ($\iota\Phi = +\Phi$).
- **QCD:** center neutrality selects confined hadronic words ($z(w) = 1$, $\varepsilon(w) = +1$).
- **Strong CP:** sheet-odd cancellation kills θ_{sheet} in the relative action.

These are not three independent rules bolted onto different sectors. They are three projections of the same admissible-versus-inadmissible boundary: seam parity, center charge, and sheet exchange are all encoded in the double-cover and involution data of the primitive core. This unification is the conceptual backbone of the transport program.

8 Particle and mass compiler

This section is the phenomenological heart of the manuscript. It is not meant to flatten all rows into one uniform notion of prediction. Instead it makes the compiler interface explicit: a declared RG map, a kernel-output map, and a mass ledger that separates genuinely derived quantities from solver-level candidates and anchor-dependent comparison rows. Accordingly this section is best read through three ledgers:

- a *primary benchmark ledger* for the narrow main-text comparison surface in Table 10,
- *mass-closure ledgers* for fermion, electroweak, and hadronic rows in Tables 6 and 7,
- and a *supplementary / anchor-dependent ledger* in the appendix for rows that remain comparison-heavy or exploratory.

8.1 RG map for compiler outputs

The symbol $\mathcal{R}_{\mathrm{SM}}$ denotes the declared comparison map from TFPT outputs to scheme-specified observables. Here it is the bookkeeping package

$$\mathcal{R}_{\mathrm{SM}} = (\mu_{\mathrm{match}},\ \{\mu_{\mathrm{thr}}\},\ \text{running convention},\ \text{scheme conversion},\ \text{observable map}),$$

and we write

$$\bar{\alpha}^{(5)}(M_Z) := \alpha_{\mathrm{em}}^{\overline{\mathrm{MS}},\, n_f=5}(M_Z).$$

with the following restricted interpretation in the main text:

- $\alpha^{-1}(0)$ is compared in the Thomson-limit on-shell convention, with the CODATA 2022 / NIST value taken as the normative low-energy reference rather than the separate PDG electroweak extraction from a_e,
- $\bar{\alpha}^{(5)}(M_Z)^{-1}$ is compared after a declared threshold map into the $\overline{\mathrm{MS}}$ language at $\mu = M_Z$,
- neutrino-angle rows use the NuFIT normal-ordering reference snapshot based on data available through November 2025,
- the Ω_b row is reconstructed from the Planck 2018 base-ΛCDM pair $(\Omega_b h^2, H_0)$ via $\Omega_b = (\Omega_b h^2)/h^2$ with $h = H_0/100$,
- the β row uses the Minami–Komatsu extraction from Planck 2018 polarization as the normative benchmark, while PR4 and PR4 map-space analyses are kept only as supplementary context,
- direct-search interface rows are not treated as precision residuals and are therefore kept out of the main benchmark table.

For later use we also set

$$\beta_{\mathrm{rad}} := \frac{\varphi_0}{4\pi}.$$

8.2 Central hard-kernel quantities

The roles of c_3 and φ_0 should not remain implicit. They are the central hard numeric quantities of the present main-text kernel. The downstream E_8 parameter κ_{E8} is isolated in the next section rather than folded back into the kernel center.

The most exposed closed formulas driven by these constants are those already displayed in Equation (12).

Quantity	Closed value	Role / status	Named outputs
c_3	$1/(8\pi)$ $= 0.039\,790\,000\,00$	Hard seam normalization	$\delta_{\rm top}$, $C_{a\gamma\gamma} = -4c_3$, $\sigma = c_3^2\chi_{\rm QCD}^2$, $M/\bar{M}_{\rm Pl} = \sqrt{8\pi}c_3^4$
φ_0	$\frac{1}{6\pi} + \frac{3}{256\pi^4}$ $= 0.053\,170\,000\,00$	Hard corrected opening; bridge reinterpretation via $\delta_{\rm top} = \Omega_{\rm adm}c_3^4$	$\beta_{\rm rad} = \varphi_0/(4\pi)$, $\lambda_C = \sqrt{\varphi_0(1-\varphi_0)}$, $\sin^2\theta_{13}$, charged-lepton ladders

Table 5: Central hard-kernel quantities and the main predictions they feed.

8.3 Mass-closure ledgers

For the mass surface itself the main text now uses only three row-level status words: *derived*, *candidate*, and *anchor-dependent*. The explicit candidate formulas, neutrino rows, and transport-ratio source data are moved to the appendix so that the main text shows a compact comparison-ready ledger rather than a mixed formula dump.

The compact ledgers are intentionally narrow: they show the comparison surface, not the internal algebra. Explicit candidate formulas for e, μ, τ, neutrino branch values, and the ladder / ratio source rows for the quark sector are collected in the appendix-level supplementary mass-source table Table 19.

9 E_8 as downstream scale grammar

E_8 is not part of the primitive carrier ontology and should not compete with the seam data, the carrier theorem, or the completion program for explanatory priority. In the present main-text form it plays one restricted role only: a downstream scale grammar attached to the already fixed carrier norm.

$$\kappa_{E8} := \frac{\gamma}{\ln(248/60)} = \frac{5/6}{\ln(248/60)}.$$

9.1 Block formula and reading discipline

A physical E_8 representation is not the stage label n alone, but the full block triple (n, r, I_1). The compressed block formula is

$$X(n, r, I_1) = \frac{M_{\rm Pl}}{8}\,\sin^2\theta_{13}\left(\frac{60-2n}{60}\right)^{\kappa_{E8}} e^{-(8-r)/64}\, e^{-12\pi I_1}, \tag{32}$$

where $M_{\rm Pl} \approx 1.220\,900\,000 \times 10^{19}$ GeV is the *unreduced* Planck mass (not $\bar{M}_{\rm Pl} = M_{\rm Pl}/\sqrt{8\pi}$), n is the ladder stage with $D_n = 60 - 2n$, r is a mild block-multiplicity label ($1 \le r \le 8$), and I_1 is the dominant block trace. The reading discipline is strict: n orders the ladder, I_1 controls most of the exponential suppression via $e^{-12\pi I_1}$, and r only refines the scale. In the relevant energy range the step ratio between successive stages is

$$\frac{X_{n+1}}{X_n} = \left(\frac{58-2n}{60-2n}\right)^{\kappa_{E8}} \approx 0.96\text{–}0.97,$$

so large hierarchies come almost entirely from I_1, not from n.

9.2 Restricted stage atlas

With the restrictive trace grammar $I_1 \in \{\frac{1}{12}, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{5}{8}, \frac{41}{48}, 1, \frac{17}{16}\}$ the ladder produces a set of robust stage identifications. The main-text table shows only the robust nodes; the full real-domain atlas up to $n = 29$—including restricted-scan candidates and IR tail conjectures—is given in Table 17 in the appendix.

Particle	TFPT source class	TFPT value	PDG value	Residual	Status
e	constant-factory candidate	0.520 960 000 0 MeV	0.510 999 000 0 MeV	+1.95%	candidate
μ	constant-factory candidate	107.490 000 0 MeV	105.658 000 0 MeV	+1.73%	candidate
τ	constant-factory candidate	1.768 800 000 GeV	1.776 860 000 GeV	−0.45%	candidate
Calibration surface (anchor-dependent; zero residual = anchor consistency, not prediction):					
$W^{\pm}$	boundary EW input at M_Z	80.369 000 00 GeV	80.369 000 00 GeV	0.00%	anchor
Z	boundary EW input at M_Z	91.188 000 00 GeV	91.188 000 00 GeV	0.00%	anchor
H	boundary Higgs input	125.250 000 0 GeV	125.250 000 0 GeV	0.00%	anchor

Table 6: Compact mass surface, electroweak and charged-lepton sector. Zero residual on anchor-dependent rows records consistency with the chosen anchor only.

Particle	TFPT source class	TFPT value	PDG value	Residual	Status
u	light-quark solver relative to strange anchor, $\overline{\mathrm{MS}}$ at 2 GeV	2.195 000 000 MeV	2.160 000 000 MeV	+1.62%	derived
d	light-quark solver relative to strange anchor, $\overline{\mathrm{MS}}$ at 2 GeV	4.670 000 000 MeV	4.670 000 000 MeV	+0.00%	derived
π^0	GMOR hadronic cross-check	134.979 000 0 MeV	134.977 000 0 MeV	+0.00%	derived
p	hadron mass-ladder cross-check	935.030 000 0 MeV	938.272 000 0 MeV	−0.35%	derived
Calibration surface (anchor-dependent):					
s	strange-quark anchor, $\overline{\mathrm{MS}}$ at 2 GeV	93.400 000 00 MeV	93.400 000 00 MeV	0.00%	anchor
c	up-type ratio ladder with threshold input	1.270 000 000 GeV	1.270 000 000 GeV	0.00%	anchor
b	down-type ratio ladder with threshold input	4.180 000 000 GeV	4.180 000 000 GeV	0.00%	anchor
t	up-type ratio ladder with threshold input	172.570 000 0 GeV	172.570 000 0 GeV	0.00%	anchor

Table 7: Compact mass surface, quark and first-hadron sector. Zero residual on anchor-dependent rows records consistency with the chosen anchor only.

n	r	I_1	X (GeV)	Target	Residual	Identification
3	8	1/2	$2.159\,000\,000 \times 10^8$	$2.160\,000\,000 \times 10^8$	0.06%	μ_{c_3} RG crossing
5	5	1/12	$1.307\,000\,000 \times 10^{15}$	$1.310\,000\,000 \times 10^{15}$	0.2%	seesaw M_R
10	1	1/3	$8.826\,000\,000 \times 10^{10}$	direct search	—	PQ scale f_a
12	2	41/48	246.350 000 0	246.220 000 0	0.05%	EW vacuum v
14	4	17/16	0.092 100 000 00	0.092 700 000 00	0.6%	pion band
15	4	1	0.935 000 000 0	0.938 000 000 0	0.3%	hadron anchor
21	4	1/6	$3.051\,000\,000 \times 10^{13}$	$3.060\,000\,000 \times 10^{13}$	0.3%	inflation M
21	5	41/48	171.800 000 0	172.600 000 0	0.4%	top quark
24	7	5/8	$7.895\,000\,000 \times 10^5$	$7.920\,000\,000 \times 10^5$	0.3%	μ_{φ_0} RG crossing
25	5	1	0.498 000 000 0	0.498 000 000 0	0.0%	kaon band
25	7	41/48	125.500 000 0	125.250 000 0	0.2%	Higgs mass
26	2	1	0.417 000 000 0	0.420 000 000 0	0.7%	$\sqrt{\sigma}$ confinement

Table 8: Robust E_8 stage atlas (restrictive trace grammar only). Restricted-scan candidates and tail conjectures are in Table 17.

Remark (Non-independence of stage-atlas entries). The stage-atlas entries are not statistically independent confirmations. They are projections of a single block grammar $X(n, r, I_1)$ with three discrete parameters. The grammar organizes scale ratios once the carrier and transport seeds are fixed, but the number of "hits" is bounded by the number of (n, r, I_1) triples that can be formed, not by the number of independent physical measurements. This is the scale-grammar analogue of the one-seed warning for the benchmark bridge rows.

9.3 Split-node structure

Two stages carry a particularly revealing double structure.

Proposition 9.1 (Inflation–top split node)**.** At $n = 21$, the E_8 block formula admits two well-separated physical identifications:

$$X(21, 4, \tfrac{1}{6}) \approx 3.051\,000\,000 \times 10^{13}\ \text{GeV} \quad \text{(inflation scale)}, \qquad X(21, 5, \tfrac{41}{48}) \approx 171.800\,000\,0\ \text{GeV} \quad \text{(top mass)}.$$

The same orbit depth carries the UV and IR endpoints of the electroweak hierarchy.

Proposition 9.2 (Higgs–kaon split node)**.** At $n = 25$, the block formula admits

$$X(25, 7, \tfrac{41}{48}) \approx 125.500\,000\,0\ \text{GeV} \quad \text{(Higgs mass)}, \qquad X(25, 5, 1) \approx 0.498\,000\,000\,0\ \text{GeV} \quad \text{(kaon band)}.$$

Again a single orbit depth carries two widely separated physical sectors, distinguished entirely by the block trace I_1.

The split-node structure shows that n is a radial orbit coordinate and I_1 is the sector projection. A physical representation of the E_8 ladder is therefore (n, r, I_1), not n alone.

9.4 Permissiveness barrier and real-domain boundary

Remark (Permissiveness barrier at $n = 18$,19). Under the restricted trace grammar, stages $n = 18$ and $n = 19$ have no unambiguous physical target. If I_1 is allowed to scan over all simple rationals, these stages can mimic almost any scale between 10^2 and 10^{16} GeV. They mark the exact boundary where E_8 without a derived block-trace theorem becomes too permissive. The open key question of the E_8 layer is therefore not the ladder itself but the derivation of $I_1(B)$ from carrier and block structure. Until that derivation is available, stages 18 and 19 are the sharpest internal test points.

Quantity	Value on the retained branch	Main-text role
κ_{E8}	0.587 230 000 0	downstream ladder exponent fixed by $\gamma = 5/6$
D_1	60	optional ladder start index; arithmetically tied to the canonical branch
f_a	$8.860\,000\,000 \times 10^{10}$ GeV	representative axion-side scale on the retained branch
m_a	65.190 000 00 μeV	representative downstream mass scale
ν_a	15.764 000 00 GHz	representative frequency target

Table 9: Short main-text E_8 parameter ledger. The role of E_8 here is organizational rather than ontological.

Obs.	TFPT formula	TFPT value	Exp. value	Unc.	Residual	Scale / scheme	Status
$\alpha^{-1}(0)$	self-consistent carrier closure $\varphi_0(\alpha)$ and δ_2	137.035 999 2	137.035 999 2	$2.100\,000\,000 \times 10^{-8}$	$+1.87\sigma$	Thomson on-shell	Hard
$\bar{\alpha}^{(5)}(M_Z)^{-1}$	$\mathcal{R}_{\rm SM}[\alpha(0)]$	127.940 500 0	127.930 000 0	0.008 000 000 000	$+1.32\sigma$	$\overline{\rm MS}$ at M_Z	Bridge
λ_C	$\sqrt{\varphi_0(1-\varphi_0)}$	0.224 380 000 0	0.224 310 000 0	$8.500\,000\,000 \times 10^{-4}$	$+0.08\sigma$	CKM proxy	Bridge
$\sin^2\theta_{13}$	$\varphi_0 e^{-5/6}$	0.023 110 000 00	0.022 240 000 00	$5.700\,000\,000 \times 10^{-4}$	$+1.524\sigma$	NuFIT NO	Bridge
Ω_b	$(4\pi-1)\beta_{\rm rad}$	0.048 940 000 00	0.049 300 000 00	$8.570\,000\,000 \times 10^{-4}$	-0.421σ	Planck 2018 from $\Omega_b h^2$	Bridge
β	$\beta_{\rm rad}$ in degrees	0.242 000 000 0	0.350 000 000 0	0.140 000 000 0	-0.77σ	Planck 2018 Minami–Komatsu	Bridge

Table 10: Primary benchmark table for the main-text claims. Supplementary rows and out-of-sample checks are collected in Appendix F.

Remark (Real-domain theorem). The hard nilpotent chain gives $D_n = 60 - 2n$ for $n = 1, \ldots, 26$. Real-valued continuation is possible up to $n = 29$ ($D_{29} = 2 > 0$). At $n = 30$, $D_n = 0$ and the power $(D_n/60)^{\kappa_{E8}}$ hits a terminal wall. For $n \geq 31$, $D_n < 0$ and the irrational exponent κ_{E8} produces complex values. The physically meaningful real-valued ladder therefore terminates at $n = 29$.

10 Primary benchmark map and prediction ledger

The comparison values used in this section follow representative CODATA, PDG, NuFIT, Planck, and birefringence references [3, 4, 5, 6, 7, 8, 9]. The main-text table is intentionally restricted to six rows whose comparison language is already reasonably convention-complete. The compact mass surface is given separately in Tables 6 and 7; more exploratory rows and the main unresolved comparison are moved to Appendix F. Here "carrier closure" abbreviates the self-consistent electromagnetic closure expression with explicit $\varphi_0(\alpha)$ and second-order term $\delta_2 = \frac{5}{4}\delta_{\rm top}^2$. The $\alpha^{-1}(0)$ row is therefore structurally different from the four bridge rows $(\beta, \Omega_b, \lambda_C, \sin^2\theta_{13})$, which are all projections of the same retained seed φ_0.

Remark (One seed, four projections). The bridge rows are not statistically independent hits. With

$$\beta_{\rm rad} = \frac{\varphi_0}{4\pi}, \qquad \Omega_b = (4\pi - 1)\beta_{\rm rad}, \qquad \lambda_C = \sqrt{\varphi_0(1-\varphi_0)}, \qquad \sin^2\theta_{13} = \varphi_0 e^{-\gamma},$$

the four-row map is a one-variable seed map

$$\mathcal{S}(\varphi_0) := (\beta_{\rm rad}, \Omega_b, \lambda_C, \sin^2\theta_{13}),$$

Domain	Status	Output package	Source layer
Carrier packet	Hard	$E_3 \oplus E_2$, $G_{\mathrm{SM}}^{\mathrm{glob}}$, S^+, γ, b_1	minimal carrier theorem and half-spinor decomposition
Kernel numerics	Hard	c_3, φ_0, α, $C_{a\gamma\gamma}$, β, λ_C	carrier-form closure equation and seam constants
Family / occupancy layer	Bridge	F, $\Omega_{\mathrm{adm}} = 48$, δ_{top}	monodromy and occupancy bridge
Geometric compiler	Completion	Φ, $\bar{M}_{\mathrm{Pl}}$, v, R^2 channel	bosonic index plus relative heat-kernel closure
Transport compiler	Completion	$\mathcal{Y}_f$, m_f, CKM, PMNS	phase-distance kernel and path-length mass closure
QCD closure	Completion	confinement, hadronic admissibility, $\bar{\theta} = 0$	center neutrality plus positive-kernel hardening
Downstream ladder	Conditional	E_8 scale ratios and inflation-side imports	optional appendix-level scale grammar
Observer layer	Conditional	record algebra, horizons, prediction semantics	appendix-level extension

Table 11: Compact prediction ledger for the carrier-first compiler presentation.

with Jacobian

$$\frac{d\mathcal{S}}{d\varphi_0} = \left(\frac{1}{4\pi},\, 1 - \frac{1}{4\pi},\, \frac{1 - 2\varphi_0}{2\sqrt{\varphi_0(1-\varphi_0)}},\, e^{-\gamma} \right).$$

Thus β, Ω_b, λ_C, and $\sin^2\theta_{13}$ are one seed and four shadows, not four detached benchmark events. They remain in the table because their comparison conventions and uncertainty models differ, not because the underlying algebraic source is independent.

10.1 Compact prediction ledger

The main benchmark table shows only a narrow comparison surface. To keep the compiler structure visible, Table 11 summarizes which output packages belong to which layer of the paper.

10.2 Residual map and principal unresolved comparisons

The residuals for the restricted six-row table are shown in Figure 6. The carrier-linked electromagnetic, birefringence, and baryon-density channels all remain within the displayed few-sigma window. The CKM phase is not plotted here because it is better treated, at present, as the main unresolved transport-side comparison and is therefore moved to Appendix F.

The principal unresolved comparisons are therefore specific rather than diffuse:

- the CKM phase needs either a genuine discrete derivation or a cleaner statement that the current relation is only a transport bridge,
- the bosonic relative index remains the decisive geometric theorem for the Higgs sector,
- the χ-channel heat-kernel closure must be shown in one declared scheme before the induced-Einstein narrative can be treated as more than a coherent target.

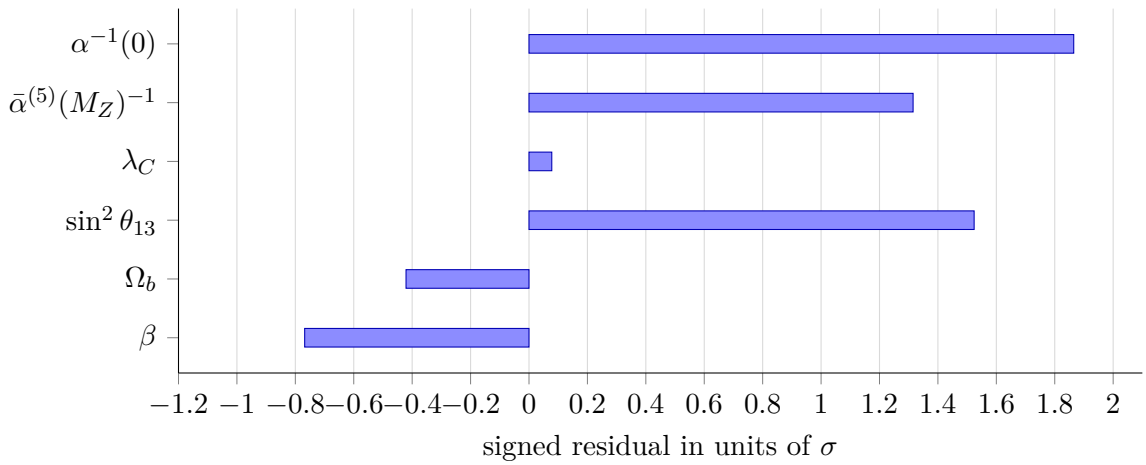

Figure 6: Residual plot for the main benchmark rows with declared uncertainties.

10.3 Falsification matrix

11 Conditional master theorem and finite proof obligations

11.1 Compiler chain

Collecting the present hard, bridge, and completion layers, the intended compiler line is

$$\begin{aligned}(\widetilde{M} \to M, \Sigma, \iota, \chi) &\Rightarrow (E_3 \oplus E_2,\ G_{\mathrm{SM}}^{\mathrm{glob}},\ S^+)\\ &\Rightarrow (N_{\mathrm{fam}} = 3,\ \Omega_{\mathrm{adm}} = 48,\ F,\ \Phi)\\ &\Rightarrow (\bar{M}_{\mathrm{Pl}},\ v,\ \mathcal{Y}_f,\ \sigma,\ \bar{\theta})\\ &\Rightarrow (m_f,\ V_{\mathrm{CKM}},\ U_{\mathrm{PMNS}},\ \text{hadrons})\\ &\Rightarrow (\text{benchmarks},\ \text{Pred}).\end{aligned} \tag{33}$$

This line is not meant as a slogan. Each arrow is attached either to a hard theorem, a bridge construction, or one of the explicit completion interfaces listed in the proof-obligation table below.

11.2 Conditional master theorem

Theorem 11.1 (Conditional master theorem)**.** *Assume the hard carrier and kernel layers summarized in Table 1, together with the completion statements of the geometric and transport programs and the declared comparison map $\mathcal{R}_{\mathrm{SM}}$. Then every observable in the present main-text scope is the output of one carrier-first relative spectral compiler from primitive seam data to scheme-specified observables.*

Proof sketch. The proof is compositional. The hard layer fixes the carrier, the global Standard Model quotient, one family, the carrier norm, the abelian index, and the established electromagnetic kernel. The bridge layer lifts the carrier packet to family multiplicity and occupancy and installs the RG/comparison interface. The completion layer is then finite: a bosonic relative index is asked to select the unique light Higgs mode, the χ channel is asked to produce $\bar{M}_{\mathrm{Pl}}$ and v, the phase-distance transport kernel is asked to generate Yukawas and mixings, and center-neutral admissibility is asked to close confinement and strong CP. Once these completion arrows are granted, no further primitive category is needed for the particle, mass, and benchmark compiler described here. □

Statement	What would falsify it	Sector affected	Local or global failure
Minimal carrier $E_3 \oplus E_2$	failure to recover the one-family packet or the global quotient from the carrier theorem	Hard	global
Family bridge and $\Omega_{\mathrm{adm}} = 48$	no geometric monodromy interpretation of the three-cycle family space	Bridge	local-to-structural
Bosonic relative index selecting Φ	no unique seam-even doublet or an unavoidable extra light triplet	Completion	local
χ-channel Einstein bridge	inability to obtain one common closure for $\bar{M}_{\mathrm{Pl}}$, v, and the comparison map	Completion	global
Transport positivity and strong CP closure	nonzero determinant phase or an unavoidable $\bar{\theta}$ counterterm	Completion	local
Confinement as admissible word selection	necessity of physical states with $z(w) \neq 1$ in the low-energy spectrum	Completion	local

Table 12: Falsification matrix for the present main-paper claims.

11.3 Finite proof obligations

12 Conclusion

The present paper argues that a small carrier-first core already fixes nontrivial physics: the minimal $3 + 2$ carrier, the global Standard Model quotient, one chiral family, the carrier norm $5/6$, the abelian index, the seam constant c_3, the corrected opening φ_0, and the electromagnetic closure channel. The compression identities and exclusion theorems then show that this core is overdetermined: the rank-lock, the 41-chain, the Cabibbo inversion, and the categorical failure of wrong discrete worlds are not additional ornaments but structural consequences of the same small algebra. What remains open is no longer a vague enlargement of the ontology, but a finite list of completion interfaces.

Those interfaces are explicit. The family local system must harden nondegeneracy and mixing; the family count itself is already supported by the local patch-deficit route. A bosonic relative index must select the unique light Higgs doublet. The χ channel must close the Planck and electroweak scales in one geometric response, with the Newton-form relation understood as a corollary rather than a second output. The transport kernel must harden Yukawas, mixings, confinement, and strong CP while separating positive radial mass data from unitary family holonomy cleanly enough that strong and weak CP are not conflated. The α row must remain tied to the full self-consistent solver rather than to a frozen opening. If those finite steps are completed, the same relative spectral framework yields one compiler from seam data to particle content, masses, hadronic admissibility, and scheme-specified observables. The open problem is therefore execution of a finite proof program, not addition of a new primitive layer.

A Technical conventions and data layers

This appendix collects the technical bookkeeping that supports the carrier-first presentation but would slow the early main-text flow if placed before the carrier theorem.

Target statement	Input data	Missing lemma	Minimal deliverable	Empirical consequence
Family space theorem	Σ, C_1, C_2, C_T, $T^3 = 1$	monodromy-to-family theorem	derive F from the seam cycle algebra	stabilizes the meaning of $\Omega_{\mathrm{adm}} = 48$
Bosonic index closure	$E_3 \oplus E_2$, $\mathcal{B}_{\mathrm{rel}}$	relative bosonic index theorem	prove unique seam-even doublet selection	Higgs-sector hardening
χ-channel closure	$\mathbb{A}_\Sigma$, APS data, heat-kernel coefficients	relative heat-kernel computation	explicit coefficient map with declared scheme	Planck / Higgs bridge becomes testable
Transport distance theorem	U_6, D_y, admissible words	phase-distance theorem	derive $L_{f,j}$ and bounded $\Lambda_{f,j}$ from transport geometry	CKM / PMNS and mass hierarchy become sharper
Comparison bridge	$\alpha(0)$, Ω_b, β, scheme map $\mathcal{R}_{\mathrm{SM}}$	RG and matching control	one declared threshold / scheme section	quantitative theory-versus-data map becomes convention-complete

Table 13: Finite proof obligations for the present main-paper form.

Relative object	Definition or schematic form	Role
D_{rel}	$D - D_{\mathrm{ref}}$ or a specified relative pair (D, D_{ref})	fermionic comparison operator
S_{rel}	$\mathrm{Tr}\, f(D/\Lambda) - \mathrm{Tr}\, f(D_{\mathrm{ref}}/\Lambda) + \frac{i\pi}{2}\Delta\eta$	relative spectral action
$\mathrm{Ind}_{\mathrm{rel}}$	APS-type index of a relative pair or superconnection package	carrier and bosonic counting
$\langle W(C)\rangle_{\mathrm{rel}}$	Wilson observable after subtraction of the declared reference sector	confinement diagnostics
$\mathcal{H}_{\mathrm{rel}}$	relative Hamiltonian used in admissible-sector minimization	optional record extension

Table 14: Conventions on relative objects.

A.1 Relative objects

The adjective "relative" is used in one fixed sense throughout the paper: every operator or action is compared to a declared reference object.

A.2 Layered datum and admissibility

Construction A.1 (Layered datum)**.** The technical bookkeeping behind the manuscript uses four levels:

$$\mathfrak{T}_{\mathrm{core}} = (\widetilde{M} \to M,\ \Sigma,\ \iota,\ D_{\mathrm{rel}},\ \mathsf{Adm}),$$

$$\mathfrak{T}_{\mathrm{ker}} = (E_3 \oplus E_2,\ Y,\ S^+,\ \gamma,\ B,\ G^{\mathrm{glob}}_{\mathrm{SM}},\ b_1(N_{\mathrm{fam}}, N_\Phi)),$$

$$\mathfrak{T}_{\mathrm{bridge}} = ((F,T),\ \Omega_{\mathrm{adm}},\ \delta_{\mathrm{top}},\ \mathcal{R}_{\mathrm{SM}}),$$

$$\mathfrak{T}_{\mathrm{comp}} = ((\chi, \mathcal{B}_{\mathrm{rel}}, \mathbb{A}_\Sigma),\ (U_6, D_y, \mathcal{Y}_y^{(\varepsilon)})).$$

Level	Objects	Claim role	Function in the present paper
Core datum	$\widetilde{M} \to M$, Σ, ι, D_{rel}, Adm	Hard	minimal input required to formulate admissibility and the carrier theorem
Carrier kernel	$E_3 \oplus E_2$, Y, S^+, γ, B, $G_{\mathrm{SM}}^{\mathrm{glob}}$, $b_1(N_{\mathrm{fam}}, N_\Phi)$	Hard	rigid particle-theory packet extracted from the core datum
Bridge layer	(F, T), Ω_{adm}, δ_{top}, $\mathcal{R}_{\mathrm{SM}}$	Bridge	organizes family lift, occupancy, and scheme-specified comparison
Completion data	$(\chi, \mathcal{B}_{\mathrm{rel}}, \mathbb{A}_\Sigma)$, $(U_6, D_y, \mathcal{Y}_y^{(\varepsilon)})$	Completion	collects the open geometric and transport interfaces without promoting target outputs to primitive data

Table 15: Core datum versus kernel, bridge, and completion structures.

Conditional record, horizon, and prediction-semantics data are kept outside this compact layered datum and are introduced only in the dedicated appendix-level extension.

Definition A.2 (Admissibility predicate)**.** In the present paper Adm denotes a predicate on relative sectors, and by mild abuse of notation also the induced class of admissible sectors. We write $\mathsf{Adm}(X) = 1$ precisely when:

(1) X lies in the declared domain of the relative operator pair,

(2) the associated relative action or index density is finite after reference subtraction,

(3) the required seam-evenness condition is satisfied,

(4) colored sectors satisfy center neutrality when a hadronic interpretation is intended,

(5) positivity and closure conditions hold in those sectors used for observable or transport statements.

Thus Adm is not introduced here as a projector or a functor, but as a predicate that defines the class of sectors on which the later carrier, transport, and appendix-level constructions are allowed to act.

B Constants atlas and downstream scale grammar

This appendix collects material that is useful for internal compression but should not dominate the main-text scope of explicit claims.

B.1 Global rotation angle and constants atlas

The seam loop is assumed to carry one unit of spectral flow,

$$U_\Sigma(\theta) = e^{i\theta}, \qquad \mathrm{SF}(U_\Sigma) = 1, \qquad \Delta_\Sigma = 2\pi.$$

Because the spin lift closes only after 4π, the normalized rotation modulus is

$$a_\theta = \frac{\theta}{4\pi}.$$

The cascade in Figure 7 is kept here as technical bookkeeping rather than as an early main-text dependency graphic.

Here the arrow $a_\theta \to c_3$ records a normalization convention internal to the present setup, not an independently proved theorem beyond the declared seam and spin-lift normalization.

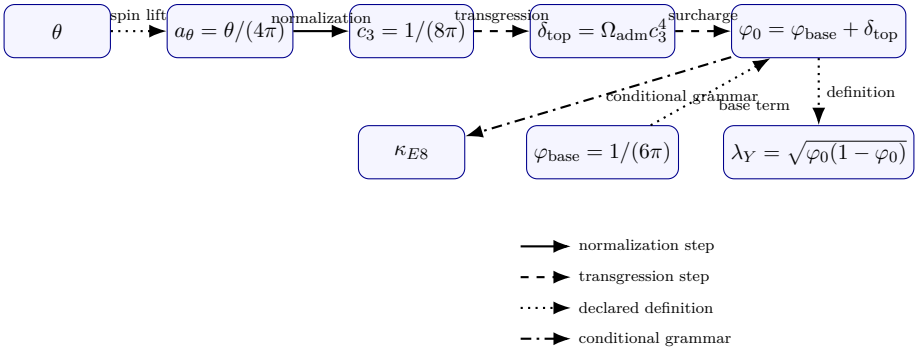

Figure 7: Constant cascade used for technical bookkeeping, shown in the occupancy-reinterpretation form for $\delta_{\rm top}$. Arrow styles encode algebraic role rather than claim status: normalization, transgression, declared definition, and conditional downstream grammar.

B.2 E_8 as downstream scale grammar

In the present paper E_8 is not used as a primitive cause. The retained parameter is

$$\kappa_{E8} := \frac{\gamma}{\ln(248/60)} = \frac{5/6}{\ln(248/60)}.$$

On the retained branch this means

$$\begin{aligned} \kappa_{E8} &\approx 0.587\,230\,000\,0, & f_a &\approx 8.860\,000\,000 \times 10^{10}\,\text{GeV}, \\ m_a &\approx 65.190\,000\,00\,\mu\text{eV}, & \nu_a &\approx 15.764\,000\,00\,\text{GHz}. \end{aligned}$$

The intended reading is a downstream scale grammar:

- the carrier packet fixes $\gamma = 5/6$,
- κ_{E8} then defines an optional ladder parameter,
- the ladder is allowed to organize scale spacing,
- but it does not replace the carrier theorem or the seam data.

In the fuller appendix-level grammar one may package a downstream block scale as

$$X(n, r, I_1) = \frac{M_{\rm Pl}}{8} \sin^2\theta_{13} \left(\frac{60-2n}{60}\right)^{\kappa_{E8}} e^{-(8-r)/64} e^{-12\pi I_1},$$

where n is the ladder stage, r a mild block-multiplicity label, and I_1 the dominant block trace. This makes the reading discipline explicit: the stage index orders the ladder, the trace controls most of the exponential suppression, and the extra block label only refines the scale. E_8 is therefore a downstream scale grammar, not a universal mass formula.

B.3 Full stage atlas

The full real-domain atlas up to $n = 29$ extends the main-text robust table with restricted-scan candidates and IR tail conjectures. Rows are classified as *robust* (restrictive grammar, unambiguous target), *restricted-scan* (plausible but less constrained), or *tail conjecture* (new trace class $I_1 \in \{\frac{11}{12}, \frac{7}{6}\}$ not yet derived from block structure).

Symbol	Definition	Compiler role	First origin	Reappears in	Physical meaning
θ	seam-loop angle	core datum	seam data	a_θ	geometric rotation variable
a_θ	$\theta/(4\pi)$	hard normalization	spin lift	c_3, phase grammar	normalized cover rotation
c_3	$1/(8\pi)$	hard kernel constant	seam normalization	$\delta_{\rm top}$, β, σ	seam normalization constant
$\varphi_{\rm base}$	$1/(6\pi)$	hard kernel constant	carrier geometry	φ_0	base opening
$\delta_{\rm top}$	$\frac{3}{256\pi^4}$; bridge reading $\Omega_{\rm adm}c_3^4$	hard kernel value with bridge occupancy reinterpretation	retained numerical kernel	φ_0, δ_2	topological surcharge
φ_0	$\frac{1}{6\pi}+\frac{3}{256\pi^4}$	hard corrected opening	retained numerical kernel	λ_Y, β, flavor	minimal nontrivial opening
γ	$\mathrm{Tr}_E\, Y^2 = 5/6$	hard kernel invariant	carrier packet	κ_{E8}, δ_2	carrier norm
B	$\mathrm{rank}\, E_3/\mathrm{rank}\, E_2 = 3/2$	hard kernel invariant	carrier packet	δ_2	carrier rank ratio
$\Omega_{\rm adm}$	$3\times 16 = 48$	bridge quantity	family lift	$\delta_{\rm top}$	admissible chiral-state count
b_1	$\frac{4}{3}N_{\rm fam}+\frac{1}{10}N_\Phi$	hard kernel output	abelian index	benchmark map	hypercharge trace coefficient
κ_{E8}	$\gamma/\ln(248/60)$	conditional grammar	Appendix B	scale ladder	optional scale spacing

Table 16: Constants atlas for the present manuscript.

n	r	I_1	X (GeV)	Target	Residual	Identification	Class
3	8	1/2	$2.159\,000\,000\times 10^8$	$2.160\,000\,000\times 10^8$	0.06%	μ_{c_3} RG crossing	robust
5	5	1/12	$1.307\,000\,000\times 10^{15}$	$1.310\,000\,000\times 10^{15}$	0.2%	seesaw M_R	robust
6	6	1	1.272 000 000	1.270 000 000	0.2%	charm band	restricted-scan
8	1	17/16	0.105 900 000 0	0.105 660 000 0	0.2%	muon band	restricted-scan
10	1	1/3	$8.826\,000\,000\times 10^{10}$	direct search	—	PQ scale f_a	robust
12	2	41/48	246.350 000 0	246.220 000 0	0.05%	EW vacuum v	robust
14	4	17/16	0.092 100 000 00	0.092 700 000 00	0.6%	pion band	robust
15	4	1	0.935 000 000 0	0.938 000 000 0	0.3%	hadron anchor	robust
21	4	1/6	$3.051\,000\,000\times 10^{13}$	$3.060\,000\,000\times 10^{13}$	0.3%	inflation M	robust
21	5	41/48	171.800 000 0	172.600 000 0	0.4%	top quark	robust
24	7	5/8	$7.895\,000\,000\times 10^5$	$7.920\,000\,000\times 10^5$	0.3%	μ_{φ_0} RG crossing	robust
25	5	1	0.498 000 000 0	0.498 000 000 0	0.0%	kaon band	robust
25	7	41/48	125.500 000 0	125.250 000 0	0.2%	Higgs mass	robust
26	2	1	0.417 000 000 0	0.420 000 000 0	0.7%	$\sqrt{\sigma}$ confinement	robust
27	6	41/48	91.600 000 00	91.190 000 00	0.4%	Z boson	restricted-scan
28	1	7/6	$5.105\,000\,000\times 10^{-4}$	$5.110\,000\,000\times 10^{-4}$	0.1%	electron	tail conjecture
29	1	11/12	4.210 000 000	4.180 000 000	0.7%	bottom quark	tail conjecture

Table 17: Full E_8 stage atlas on the canonical branch up to the real-domain boundary $n = 29$.

B.4 Compression identities on the minimal branch

The main-text compression identities are developed in the dedicated section on compression identities and carrier exclusion. This appendix subsection collects additional arithmetic continuations and the compression chain summary.

Remark (Abelian package from occupancy and carrier norm). On the canonical branch,

$$\Omega_{\rm adm}\gamma = 48\cdot\frac{5}{6} = 40.$$

Thus the characteristic coefficient 41 is compressed to

$$41 = \Omega_{\mathrm{adm}}\gamma + 1 = 40 + 1,$$

that is, fermionic occupancy trace plus one light Higgs doublet contribution. Writing $b_Y := 5b_1/3$ for the usual hypercharge-normalized coefficient and using the block-index relation $k_B = \frac{3}{2}I_1(B)$ to define $I_1^{\mathrm{EW}} := \frac{5}{24}b_1$ and $k_{\mathrm{EW}} := \frac{5}{16}b_1$, one obtains the unified chain

$$48\, I_1^{\mathrm{EW}} = 32\, k_{\mathrm{EW}} = 10\, b_1 = \Omega_{\mathrm{adm}}\gamma + 1 = 41. \tag{34}$$

Equivalently,

$$I_1^{\mathrm{EW}} = \frac{41}{48}, \qquad k_{\mathrm{EW}} = \frac{41}{32}, \qquad b_1 = \frac{41}{10}, \qquad b_Y = \frac{41}{6}.$$

This is an algebraic signature rather than an accidental coincidence: the same integer 41 appears simultaneously as electroweak block trace, E_8 exponent, UV hypercharge trace, and carrier occupancy times norm plus one.

Remark (Rank lock and single-seed compression: main-text reference). The rank-lock identity $2^{g-3} = g - 1 \Rightarrow g = 5$ is proved as Proposition 5.1 in the main text. The single-seed compression and the Cabibbo inversion $\varphi_0 = \frac{1}{2}(1 - \sqrt{1 - 4\lambda_C^2})$ with all derived inter-observable relations are given in Equations (15) and (16). The carrier exclusion theorems for wrong discrete worlds are in Theorem 5.2.

B.5 Optional arithmetic continuations

The next relations mix the main-text carrier with optional appendix-level E_8, inflation, or geometric notation. They are useful compression patterns, but their interpretation should remain conditional or conjectural.

Remark (60, 48, and 5/4 on the optional ladder). If the optional scale ladder is started at $D_1 = 60$, then

$$\frac{D_1}{\Omega_{\mathrm{adm}}} = \frac{60}{48} = \frac{5}{4} = B\gamma.$$

On the minimal branch one may also write the same number as

$$\frac{5}{4} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}} - 1} \qquad (g_{\mathrm{car}} = 5).$$

This compresses the E_8 start index, the admissible chiral count, the mirror factor, and the two-defect coefficient to the same rational factor.

Remark (Factorization of the two-defect coefficient). From the main-text identity

$$\delta_2 = \frac{5}{4}\,\delta_{\mathrm{top}}^2 \qquad \text{and} \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4$$

one obtains

$$\delta_2 = 2880\, c_3^8.$$

On the canonical branch this factor admits the exact arithmetic decompositions

$$2880 = D_{\mathrm{start}}\,\Omega_{\mathrm{adm}} = 60 \cdot 48, \qquad 2880 = |R(E_8)|\,\dim \mathfrak{g}_{\mathrm{SM}} = 240 \cdot 12.$$

The arithmetic identity is exact; the interpretation of this coincidence is still appendix-level.

Conjecture B.1 (E_8 root-count reading)**.** On the same optional branch,

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5\,\Omega_{\mathrm{adm}}.$$

This suggests that the E_8 root count may be read as a fivefold unfolding of the admissible chiral packet, although that interpretation is not used as a main-text theorem.

Remark (Reactor seed of the optional scale ladder). If one writes the optional ladder as

$$\varphi_n = \varphi_0 e^{-\gamma} \left(\frac{D_n}{D_1}\right)^{\kappa_{E8}}, \qquad D_n = 60 - 2n,$$

then the start value is precisely the reactor-angle seed,

$$\varphi_n = \sin^2\theta_{13} \left(\frac{D_n}{D_1}\right)^{\kappa_{E8}}.$$

Thus the appendix-level ladder may be read as a discrete continuation of the same seed that controls θ_{13}.

Remark (Conditional inflation-side compression). If one imports the v2.8 Starobinsky-side relations

$$A_s = \frac{N^2}{24\pi^2}\left(\frac{M}{\bar{M}_{\rm Pl}}\right)^2, \qquad \frac{M}{\bar{M}_{\rm Pl}} = \sqrt{8\pi}\, c_3^4,$$

then on the minimal branch

$$A_s = \frac{N^2}{3\pi} c_3^8 = \frac{N^2}{8640\pi}\,\delta_2,$$

and also

$$\frac{M}{\bar{M}_{\rm Pl}} = \frac{\sqrt{8\pi}}{\Omega_{\rm adm}}\,\delta_{\rm top}.$$

This is recorded only as an appendix-level compression identity because inflation is outside the present main-text claim surface.

Remark (Dyonic intercept and the α kernel). From the dyonic appendix the local astrophysical β function reads

$$\beta_{\rm BH}(r) = \frac{Q_e Q_m}{256\pi^4 r^2} = \frac{\delta_{\rm top}}{3}\frac{Q_e Q_m}{r^2} = 16 c_3^4 \frac{Q_e Q_m}{r^2}.$$

This is a cross-link between the EHT intercept channel and the α kernel: the same topological coefficient $\delta_{\rm top} = 48 c_3^4$ that pushes α into the precision zone also controls the local dyonic β function. The local and cosmological sectors are therefore not two loose motifs but the same coefficient in two projections.

Remark (Conditional occupancy route to gravity). If the relative Einstein term is of the form

$$S_{\rm EH}^{\rm rel} = \frac{\chi^2}{192\pi^2}\,{\rm Tr}_{\mathcal{H}_{\rm ch}}\,\mathbf{1}\int d^4x\sqrt{-g}\,R,$$

and if the family bridge identifies

$${\rm Tr}_{\mathcal{H}_{\rm ch}}\,\mathbf{1} = \Omega_{\rm adm} = 48,$$

then

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_0^2}{2\pi^2}.$$

This is the occupancy-to-gravity route alluded to in the main text.

Remark (Compression chain). The appendix-level compression pattern may therefore be summarized as

$$Y \leftrightarrow \iota, \qquad \gamma = \frac{5}{6}, \qquad B\gamma = \frac{5}{4}, \qquad \frac{D_1}{\Omega_{\rm adm}} = \frac{5}{4}, \qquad |R(E_8)| = 5\,\Omega_{\rm adm},$$

$$b_1 = \frac{\Omega_{\rm adm}\gamma + 1}{10}, \qquad \Omega_b + \beta_{\rm rad} = \varphi_0, \qquad A_s = \frac{N^2}{8640\pi}\,\delta_2 \quad \text{(conditional v2.8 import)}.$$

B.6 Conditional minimal-parameter picture

The next package is stronger but also more conditional. It assumes that the carrier rank, family multiplicity, and seam normalization are all read as closed rather than only partially inherited.

Principle B.2 (Conditional minimal-parameter compression)**.** Assume the canonical branch

$$g_{\mathrm{car}} := \operatorname{rank} E = 5, \qquad N_{\mathrm{fam}} = 3, \qquad c_3 = \frac{1}{8\pi}.$$

Then the numerical kernel may be rewritten almost entirely in terms of

$$(\pi, g_{\mathrm{car}}, N_{\mathrm{fam}}) = (\pi, 5, 3).$$

Indeed, on this branch one has

$$\gamma = \frac{5}{6} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}}+1}, \qquad \dim S^+ = 2^{g_{\mathrm{car}}-1}, \qquad \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} 2^{g_{\mathrm{car}}-1} = 48,$$

$$\varphi_{\mathrm{base}} = \frac{1-\gamma}{\pi} = \frac{1}{(g_{\mathrm{car}}+1)\pi}, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4, \qquad \delta_2 = D_{\mathrm{start}}\, \Omega_{\mathrm{adm}}\, c_3^8,$$

with

$$D_{\mathrm{start}} = g_{\mathrm{car}} \dim \mathfrak{g}_{\mathrm{SM}} = 5 \cdot 12 = 60, \qquad \Omega_{\mathrm{adm}} = (g_{\mathrm{car}} - 1) \dim \mathfrak{g}_{\mathrm{SM}} = 4 \cdot 12 = 48.$$

Remark (Compressed dependency chain)*.* Under the same conditional closure, the appendix-level numerical chain reads

$$(\widetilde{M} \to M, \Sigma, \iota, \chi) \Longrightarrow E_3 \oplus E_2 \Longrightarrow Y \leftrightarrow \iota \Longrightarrow S(U(3) \times U(2)) \Longrightarrow S^+,$$

$$\Longrightarrow N_{\mathrm{fam}} = 3 \Longrightarrow \Omega_{\mathrm{adm}} = 48 \Longrightarrow (\delta_{\mathrm{top}}, \delta_2, \varphi_0) \Longrightarrow (\alpha, \lambda_C, \beta_{\mathrm{rad}}, \Omega_b, \sin^2\theta_{13}),$$

with the optional E_8, inflationary, and gravity-side sectors then read as downstream continuations of the same compressed core.

C Relative APS and superconnection setup

The geometric completion uses APS-type relative data [1]. In practice this means:

- one works with a declared pair (D, D_{ref}) rather than a single unanchored operator,
- boundary conditions on the seam Σ are part of the definition of the relative index,
- the η-term keeps track of the residual spectral asymmetry,
- the superconnection package is used in the same spirit as the spectral-action tradition [2], but here always with an explicit reference subtraction.

The appendix is intentionally short. Its role is to make the common reference structure visible without pretending that the full heat-kernel computation has already been completed.

D Yukawa kernels and positivity lemmas

The transport program is based on the regularized kernel

$$\mathcal{Y}_f^{(\varepsilon)} = (D_f^\dagger D_f + \varepsilon^2)^{-1}.$$

Three remarks are important.

1. The regulator ε is not optional: without it the kernel becomes ill-defined exactly where the admissibility question is most delicate.
2. Positivity of the kernel is a transport-side selection principle, not yet a proved theorem of the hard kernel.
3. Residual prefactors $\Lambda_{f,j}$ are bounded transport data that still require an operator-level closure. They must remain visible and must not be rhetorically hidden as if the entire mass map were already derived without remainder.

Remark (Positive-kernel determinant route to strong CP). If the physical mass map is written in the polar form

$$M_f = U_{f,L} H_f U_{f,R}^\dagger, \qquad H_f > 0, \qquad U_{f,L}, U_{f,R} \in SU(3)_F,$$

with H_f induced by the positive transport kernel $\mathcal{Y}_f^{(\varepsilon)}$, then the induced determinant phase vanishes:

$$\arg\det M_f = 0.$$

At the same time weak CP may survive through

$$V_{\mathrm{CKM}} = U_{u,L}^\dagger U_{d,L}.$$

Combined with the sheet-odd cancellation of θ_{sheet} in the relative action, this is the algebraic core of the conditional strong-CP route summarized in the main text.

E Optional record algebra, horizons, and prediction semantics

This appendix keeps the record-level constructions available without letting them dominate the scope of explicit claims in the main paper.

E.1 Minimal admissible vacuum

At finite volume or with a declared cutoff, an admissible vacuum candidate is written as

$$\rho_B := \arg\min_{\rho \in \mathcal{V}_{\mathrm{adm}}} \mathrm{Tr}(\rho \mathcal{H}_{\mathrm{rel}}),$$

provided the relative Hamiltonian is lower bounded on the admissible domain. In the absence of a full existence theorem this should be read as a minimizing-sequence statement, not as a solved cosmological theorem.

E.2 Record growth principle

The optional record algebra is

$$\mathcal{A}_{\mathrm{rec}}(t) := \bigoplus_{\Delta_i > H_{\mathrm{hub}}(t)} \mathcal{A}_i, \qquad S_{\mathrm{rec}}(t) = \log\dim \mathcal{A}_{\mathrm{rec}}(t).$$

As $H_{\mathrm{hub}}(t)$ decreases, more stable channels open. The intended arrow-of-time claim is therefore

$$\frac{dS_{\mathrm{rec}}}{dt} > 0,$$

but only as a record-growth principle, not as a replacement for detailed nonequilibrium dynamics.

E.3 Observer algebra and Born-type evaluation

An observer is modeled as a stable pointer subalgebra

$$\mathcal{A}_{\text{obs}} \subset \mathcal{A}_{\text{rec}}.$$

With orthogonal projectors $P_i \in \mathcal{A}_{\text{obs}}$ one writes the measurement channel

$$\mathcal{M}_O(\rho) = \sum_i (P_i \rho P_i) \otimes |i\rangle\langle i|.$$

In finite-dimensional sectors, positivity, additivity on orthogonal decompositions, and phase invariance lead to the standard Born-type assignment

$$p_i = \text{Tr}(\rho P_i).$$

The claim is therefore only a constrained evaluation principle on admissible sectors, not a new theorem about measurement in all settings.

E.4 Horizons as conjectural inner seams

If a horizon is modeled as an inner seam, the manuscript uses only an approximate factorization language,

$$\mathcal{H}_{\text{out}} \otimes \mathcal{H}_H \otimes \mathcal{H}_{\text{in}},$$

or, better, an algebraic split on an appropriate code subspace. This appendix intentionally avoids selling the factorization as exact fundamental structure.

E.5 Prediction semantics

The appendix-level prediction map is

$$Q = (\rho_0, \mathcal{O}_Q, \Delta_Q, t, \mathcal{C}_Q, \mathcal{R}_Q), \qquad \text{Pred}(Q) = \mathcal{R}_Q(\text{Tr}(\rho_t E_Q(\Delta)), \mathcal{C}_Q)\,,$$

with

$$\rho_t = U(t)\rho_0 U^\dagger(t).$$

This is kept only as an optional formal extension.

F Extended benchmarks

F.1 Extended comparison rows and supplementary checks

Observable	TFPT value	Reference value	Role	Status
δ_{CKM}	1.198 000 000 rad	1.196 000 000 rad	unresolved transport comparison	appendix comparison row
m_{π^0}	134.979 000 0 MeV	134.977 000 0 MeV	hadronic out-of-sample check	appendix comparison row
m_{bb}	0.001 500 000 000 eV	$< 0.200\,000\,000\,0$ eV	neutrinoless double-beta interface	appendix interface
η_B	$5.970\,000\,000 \times 10^{-10}$	$6.130\,000\,000 \times 10^{-10}$	leptogenesis-side interface	appendix interface
m_a	65.200 000 00 μeV	direct-search target	axion mass interface	appendix interface
ν_a	15.760 000 00 GHz	direct-search band	haloscope target frequency	appendix interface

Table 18: Extended comparison rows kept out of the main text.

F.2 Supplementary mass-source rows

The compact main-text mass ledgers intentionally suppress explicit formula detail. The source rows collected here retain that information without forcing the main text to read like a mixed formula notebook.

Particle	TFPT formula or source row	TFPT value	Status note
e	$\frac{v}{\sqrt{2}}\,\frac{16\pi}{7}\varphi_0^5$	0.520 960 000 0 MeV	constant-factory candidate
μ	$\frac{v}{\sqrt{2}}\,\frac{4\pi}{3}\varphi_0^3$	107.490 000 0 MeV	constant-factory candidate
τ	$\frac{v}{\sqrt{2}}\,\frac{7\pi}{6}\varphi_0^2$	1.768 800 000 GeV	constant-factory candidate
ν_1	minimal normal-ordering branch	0	oscillation-inversion branch
ν_2	$\sqrt{\Delta m_{21}^2}$	$8.614\,000\,000 \times 10^{-3}$ eV	derived from oscillation inversion
ν_3	$\sqrt{\Delta m_{31}^2}$ with seesaw cross-check	$5.015\,000\,000 \times 10^{-2}$ eV	seesaw cross-check gives $4.616\,000\,000 \times 10^{-2}$ eV
u	ChPT ratio inversion anchored to m_s	2.195 000 000 MeV	derived; $\overline{\mathrm{MS}}$ at 2 GeV
d	$m_s/(m_s/m_d)$	4.670 000 000 MeV	derived; $\overline{\mathrm{MS}}$ at 2 GeV
s	strange-quark anchor in the light-quark solver	93.400 000 00 MeV	anchor-dependent; $\overline{\mathrm{MS}}$ at 2 GeV
c	up-type ratio ladder with $m_c/m_u = 486.920\,000\,0$	1.270 000 000 GeV	threshold input fixes the absolute scale
b	down-type ratio ladder with $m_b/m_s = 43.790\,000\,00$	4.180 000 000 GeV	threshold input fixes the absolute scale
t	up-type ratio ladder with $m_t/m_c = 133.010\,000\,0$	172.570 000 0 GeV	pole / $\overline{\mathrm{MS}}$ distinction remains external
π^0	GMOR relation from $(m_u + m_d)$ and $\langle\bar{q}q\rangle$	134.979 000 0 MeV	hadronic out-of-sample check
p	hadron mass ladder	935.030 000 0 MeV	derived hadronic scale check

Table 19: Supplementary mass-source rows referenced by the compact main-text mass surface.

For the main-text baryon-fraction comparison we use the Planck 2018 base-ΛCDM conversion

$$\Omega_b = \frac{\Omega_b h^2}{h^2}, \qquad h = \frac{H_0}{100},$$

so the quoted Ω_b row is not a free-floating cosmology number but an explicit reconstruction from the Planck parameter pair $(\Omega_b h^2, H_0)$ adopted in the suite ledger.

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